

HiSIM_SOI 1.5.0 User's Manual

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1 Definition of the Device Parameters

HiSIM_SOI is valid for all possible SOI-MOSFET structures, causing different electrical conditions. The definition of the node potentials are:

- V_s : reference
- V_{gs} : gate to source
- V_{ds} : drain to source
- V_{es} : substrate contact to source
- V_{bcs} : body contact to source (automatically clamped for fully-depleted (FD) condition)
- V'_{bs} : set to zero for floating-body (FB) devices and to V_{bcs} for body-contact (BC or body-tie) devices

The internal node potential is denoted by V_{bs} , which is equal to $\phi_{b,SOI}$ for the floating body and to V_{bcs} for the body tie.

The model branches are summarized in Fig. 1.

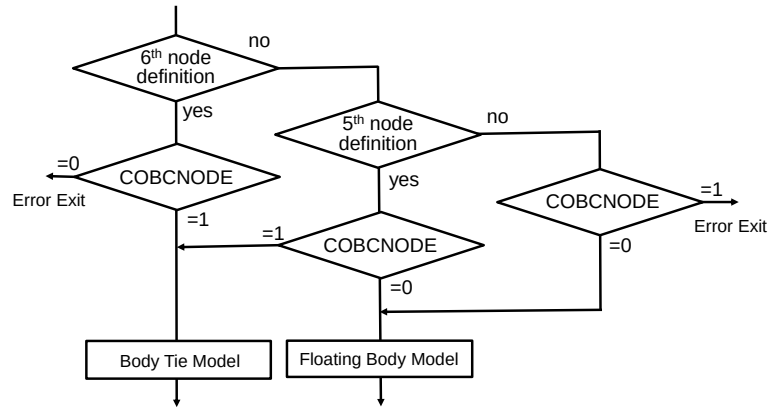


Fig. 1: Model branches considered in HiSIM_SOI.

The dynamical-depletion condition during switching is also automatically done within the model framework. The impurity concentration difference between the SOI layer **NSUBS** and the substrate **NSUBB** determines the built-in potential difference, which is also considered.

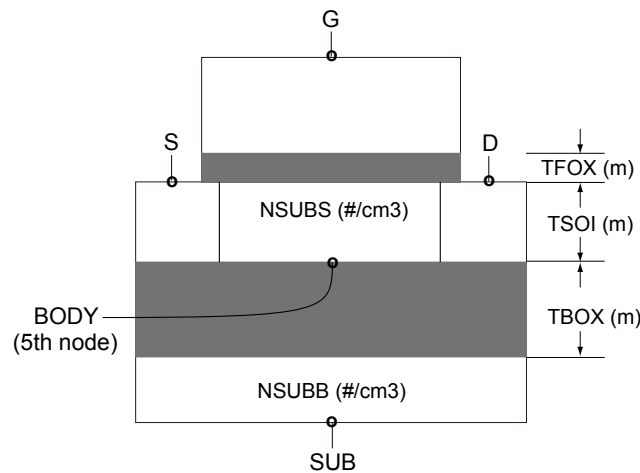


Fig. 2: Definitions of device parameters in HiSIM_SOI. The y-axis is defined along the direction of current flow (source is $y=0$) and x is defined downwards into the channel (front gate MOS interface is $x=0$).

If TBOX is so thin that the inversion condition occurs even on the backside at BOX, HiSIM_SOTB is recommended to apply. HiSIM_SOI solves the Poisson equation iteratively from the surface to the bulk substrate [1–3]. The surface

potential value, the solution of the Poisson equation, determines the device condition either fully depleted (FD) or partially depleted (PD).

Determination of the node sequence is

Drain = 1, Gate = 2, Source = 3, Substrate = 4

If COBCNODE = 1 Body = 5, Thermal = 6

If COBCNODE = 0 Thermal = 5

The effective channel length L_{eff} and width W_{eff} are calculated from the gate length L_{gate} and width W_{gate} , where L_{gate} and W_{gate} are related to the drawn gate length L_{drawn} and width W_{drawn} as

$$L_{\text{gate}} = L_{\text{drawn}} \quad (1)$$

$$W_{\text{gate}} = \frac{W_{\text{drawn}}}{\text{NF}} \quad (2)$$

$$L_{\text{eff}} = L_{\text{gate}} - 2 \cdot \text{XLD} \quad (3)$$

$$L_{\text{effc}} = L_{\text{gate}} - 2 \cdot \text{XLDC} \quad (4)$$

$$W_{\text{eff}} = W_{\text{gate}} - 2 \cdot \text{XWD} \quad (5)$$

$$W_{\text{effc}} = W_{\text{gate}} - 2 \cdot \text{XWDC} \quad (6)$$

where **XLD** and **XWD** account for the overlaps of source/drain contact and the gate oxide. The model parameter **XLDC** is introduced to describe the different length dependence of capacitances from currents. If the value is not given, the same value as **XLD** is taken. The model parameter **XWDC** is introduced to describe the different width dependence of capacitances from currents. If the value is not given, the same value as **XWD** is taken. **NF** is the number of fingers.

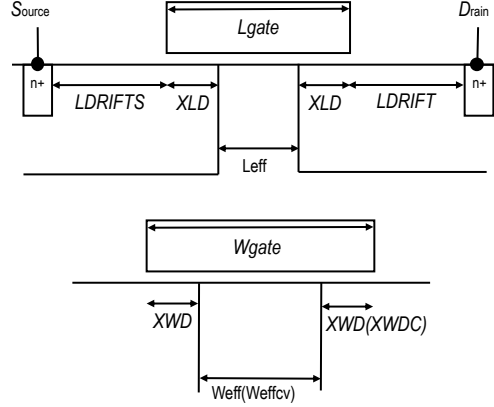


Fig. 3: Cross section of the device structure.

Table 1: HiSIM_SOI model parameters introduced in Section 1 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NF	number of gate fingers
XLD	gate overlap length
LDRIFT	length of drift region (drain side)
LDRIFTS	length of drift region (source side)
XWD	gate overlap width
XLDC	different length dependence of capacitance from currents
XWDC	different width dependence of capacitance from currents

2 Basic Equations Solved in the HiSIM_SOI Core Part

Though all following descriptions are written for the n-channel MOSFET, they are also valid for the p-channel case. The Poisson equation solved in HiSIM_SOI together with the Gauss law and boundary conditions together with the approximation of the linearly decreasing carrier density to the depth direction are

$$\phi_{s,SOI} = V'_G + \frac{Q_{s,bulk} + Q_i + Q_{dep,SOI} + Q_h}{C_{FOX}} \quad (7)$$

$$\phi_{b,SOI} = \phi_{s,SOI} + \frac{Q_{s,bulk} + \frac{1}{2}Q_{dep,SOI}}{C_{SOI}} \quad (8)$$

$$\phi_{s,bulk} = \phi_{b,SOI} + \frac{Q_{s,bulk}}{C_{BOX}} \quad (9)$$

where Q 's are charges induced within the SOI-MOSFET as depicted in Fig. 4 and Q_h is the excess charge in the SOI layer due to the impact ionization, and

$$V'_G = V_{gs} - \mathbf{VFBC} + \Delta V_{th} \quad (10)$$

$$C_{FOX} = \frac{\epsilon_{ox}}{\mathbf{TFOX}} \quad (11)$$

$$C_{BOX} = \frac{\epsilon_{ox}}{\mathbf{TBOX}} \quad (12)$$

$$C_{SOI} = \frac{\epsilon_{Si}}{\mathbf{TSOI}} \quad (13)$$

where $\mathbf{VFBC}$ is the flat-band voltage, $\mathbf{TFOX}$ is the physical gate-oxide thickness, and ΔV_{th} is the threshold voltage shift in comparison to the threshold voltage of a long-channel transistor [4]. $\epsilon_{ox}(= \epsilon_0 \cdot \kappa_{SiO_2})$ is the permittivity of SiO_2 . ϵ_0 and κ_{SiO_2} are the permittivity of vacuum and the dielectric constant of SiO_2 , respectively.

Eqs. (7)–(9) are solved simultaneously within HiSIM_SOI. For the body-tied case, the potential node $\phi_{b,SOI}$ is connected to the external node potential and solved by the circuit simulator. The solution takes place at source side and drain side. Prior to the Newton iteration, an approximated, analytical initial value for each variable is estimated in the code.

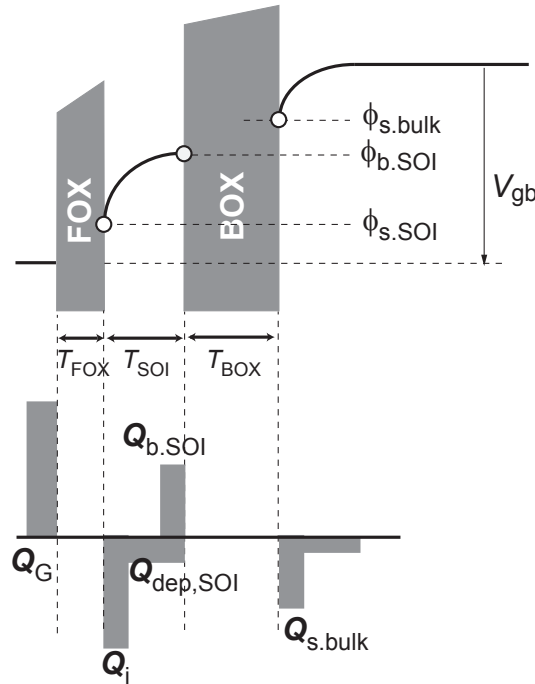


Fig. 4: Band structure of SOI-MOSFET and generated charges.

All device characteristics are determined on the basis of the charge control by applied voltages. All charges induced within SOI-MOSFET are functions of surface potentials written as

$$-Q_{\text{dep,SOI}}(y) = Q_{\text{s,dep}} + Q_{\text{b,dep}} \quad (14)$$

$$= qN_{\text{subs}}T_{\text{SOI}} : \text{ fully depleted;} \quad (15)$$

$$= \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\exp\{-\beta(\phi_{\text{s,SOI}}(y) - \phi_{\text{b,SOI}}(y))\} + \beta(\phi_{\text{s,SOI}}(y) - \phi_{\text{b,SOI}}(y)) - 1 \right]^{\frac{1}{2}} : \text{ FB partially depleted} \quad (16)$$

$$= \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\exp\{-\beta(\phi_{\text{s,SOI}}(y) - V_{\text{bcs}})\} + \beta(\phi_{\text{s,SOI}}(y) - V_{\text{bcs}}) - 1 \right]^{\frac{1}{2}} : \text{ BC partially depleted} \quad (17)$$

$$(18)$$

$$-Q_{\text{i}}(y) = \sqrt{Q_{\text{dep,SOI}}^2 + \frac{2q\epsilon_{\text{Si}}}{\beta} \exp(\beta\phi_{\text{s,SOI}})} - Q_{\text{dep,SOI}} \quad (19)$$

$$-Q_{\text{s,bulk}}(y) = \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subb}}}{\beta}} \left[\exp\{-\beta(\phi_{\text{s,bulk}}(y) - V_{\text{es}})\} + \beta(\phi_{\text{s,bulk}}(y) - V_{\text{es}}) - 1 \right. \\ \left. + \frac{n_{\text{p0}}}{p_{\text{p0}}} \left\{ \exp(\beta(\phi_{\text{s,bulk}}(y) - \phi_{\text{f}}(y))) - \exp(\beta(V_{\text{es}} - \phi_{\text{f}}(y))) \right\} \right]^{\frac{1}{2}}$$

For later reference in this manual, the factors in the above equations will be represented as

$$\text{const0} = \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \quad (20)$$

$$\text{constb} = \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subb}}}{\beta}} \quad (21)$$

$$\beta = \frac{q}{kT} \quad (22)$$

where q is the electron charge. The Boltzmann constant and the lattice temperature in Kelvin are k and T , respectively. The electron quasi-Fermi potential $\phi_{\text{f}}(y)$ preserves the following relationship:

$$\phi_{\text{f}}(L_{\text{eff}}) - \phi_{\text{f}}(0) = V_{\text{ds,eff}} \quad (23)$$

where $V_{\text{ds,eff}}$ is introduced to fit measured transition characteristics of the channel conductance g_{ds} between the linear region and the saturation region to compensate the charge-sheet approximation

$$V_{\text{ds,eff}} = \frac{V_{\text{ds}}}{\left[1 + \left(\frac{V_{\text{ds}}}{V_{\text{ds,sat}}} \right)^{\Delta} \right]^{\frac{1}{\Delta}}} \quad (24)$$

where the exponent Δ is

$$\Delta = \frac{\text{DDLTMAX} \cdot T1}{\text{DDLTMAX} + T1} + \text{DDLTICT} \quad (25)$$

$$T1 = \text{DDLTSPL} \cdot L_{\text{gate}} \times 10^6. \quad (26)$$

$V_{\text{ds,sat}}$ in (24) is written

$$V_{\text{ds,sat}} = V_{\text{g}}' + \frac{qN_{\text{subs}}\epsilon_{\text{Si}}}{C_{\text{FOX}}^2} \left\{ 1 - \sqrt{1 + 2 \frac{C_{\text{FOX}}^2}{qN_{\text{subs}}\epsilon_{\text{Si}}} \left\{ V_{\text{g}}' - \frac{1}{\beta} \right\}} \right\} \quad (27)$$

The electron concentration at equilibrium condition n_{p0} is

$$n_{p0} = \frac{n_i^2}{p_{p0}} \quad (28)$$

where n_i is the intrinsic carrier concentration and p_{p0} is the equilibrium concentration of majority carrier in the substrate, particularly, hole for n-MOSFETs, and takes an individual value in the SOI and the bulk substrate, respectively. The intrinsic carrier concentration n_i is

$$n_i = n_{i0} T^{\frac{3}{2}} \exp\left(-\frac{E_g}{2q}\beta\right) \quad (29)$$

where E_g is the bandgap.

The surface potentials, $\phi_{s0,SOI} = \phi_{s,SOI}(0)$ at the source side and at the pinch-off point $\phi_{sL,SOI} = \phi_{s,SOI}(L_{eff} - \Delta L)$ are calculated iteratively, where ΔL is the length of the pinch-off region under the saturation condition.

Different from other HiSIM models the integration of distributed charges along the channel for the FD case is done by approximating trapezoidal charge distributions as

$$Q = \frac{Q(0) + Q(L)}{2} \quad (30)$$

where 0 refers to the source side and L refers to the drain side. For the PD case, equations for the charges are derived by integrating Eq. (16) and (19). This trapezoidal equation applies for internal charges such as $Q_{b,SOI}$, $Q_{dep,SOI}$ and others wherever necessary.

Charges and currents induced within the SOI-MOSFET are connected to nodes as shown in Fig. 5 for the FB case.

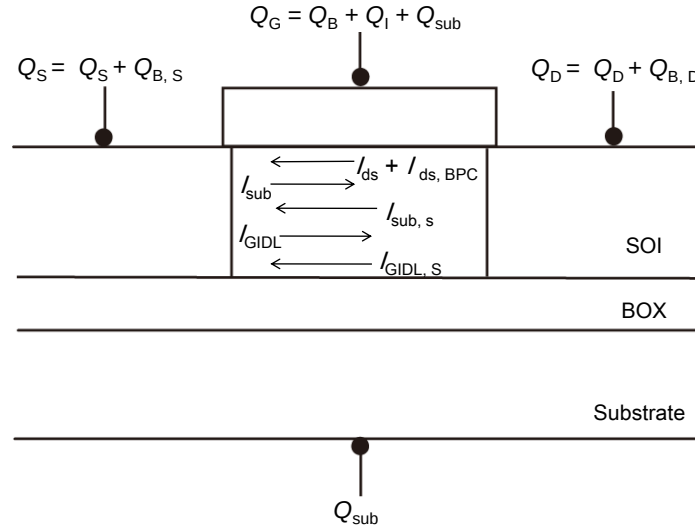


Fig. 5: Charges and currents induced within SOI-MOSFET and connections to nodes shown for the FB case.

Table 2: HiSIM_SOI model parameters introduced in Section 2 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

TFOX	front oxide thickness
TBOX	buried oxide thickness
TSOI	silicon film thickness
DDLTMAX	smoothing coefficient for V_{ds}
DDLTICT	L_{gate} -dependence of smoothing coefficient
DDLTSLP	L_{gate} -dependence of smoothing coefficient

3 Drain Current

Under the gradual-channel approximation with further approximations of an idealized gate structure and uniform channel doping, the drain current I_{ds} is written [5, 6] I_{ds} as [7] for the FB case

$$I_{ds} = \frac{W_{eff} \cdot \mathbf{NF}}{(L_{eff} - \Delta L)} \cdot \mu \cdot \frac{I_{dd}}{\beta} \quad (31)$$

$$(32)$$

$$I_{dd} = Q_{i(L)} - Q_{i(0)} - \left[\beta \cdot \frac{(Q_{i(L)} + Q_{i(0)}) \cdot (\phi_{sL} - \phi_{s0})}{2} \right] \quad (33)$$

$$(34)$$

where $Q_{i(L)}$ and $Q_{i(0)}$ are inversion charges at the drain side and the source side, respectively.

For the BT case the conventional MOSFET equation is applied, and is written

$$\begin{aligned} I_{dd} = & C_{FOX}(\beta V_G' + 1)(\phi_{sL,SOI} - \phi_{s0,SOI}) - \frac{\beta}{2} C_{FOX}(\phi_{sL,SOI}^2 - \phi_{s0,SOI}^2) \\ & - \frac{2}{3} \text{const0} \left[\left\{ \beta(\phi_{sL,SOI} - V_{bcs}) - 1 \right\}^{\frac{3}{2}} - \left\{ \beta(\phi_{s0,SOI} - V_{bcs}) - 1 \right\}^{\frac{3}{2}} \right] \\ & + \text{const0} \left[\left\{ \beta(\phi_{sL,SOI} - V_{bcs}) - 1 \right\}^{\frac{1}{2}} - \left\{ \beta(\phi_{s0,SOI} - V_{bcs}) - 1 \right\}^{\frac{1}{2}} \right] \end{aligned} \quad (35)$$

where

$$C_{FOX} = \frac{\epsilon_{ox}}{\mathbf{TFOX}} \quad (36)$$

The modeling details about the channel-length modulation ΔL are explained in the section 9.

In this section, no model parameters have been introduced.

4 Threshold Voltage Shift

The surface-potential-based model requires no threshold voltage V_{th} in its core description of the model. However, HiSIM_SOI has opted for modeling various submodels such as short-channel effects as a V_{th} shift from an otherwise ideal case.

The whole contribution to the ΔV_{th} is summarised as

$$\Delta V_{th} = \Delta V_{th,SC} + \Delta V_{th,R} + \Delta V_{th,P} + \Delta V_{th,SCR} + \Delta V_{th,W} + \Delta V_{th,sm} - \phi_{Spg} \quad (37)$$

where each term denotes:

- $\Delta V_{th,SC}$ denotes the contribution of the short-channel effects (equation 38, Section 4.1)
- $\Delta V_{th,R}$ denotes the contribution of additional short-channel effects in the presence of impurity concentration inhomogeneity in the vertical direction (retrograded implantation) (Section 4.2)
- $\Delta V_{th,P}$ denotes the contribution of the short-channel effects in the presence of impurity concentration inhomogeneity in the lateral direction (pocket implantation) (equation 43, Section 4.1)
- $\Delta V_{th,SCR}$ denotes the contribution of the short-channel effects which is specific to SOI devices (equation 42, Section 4.1)
- $\Delta V_{th,W}$ denotes the contribution of the narrow-channel effects (equation 97, Section 10)
- $\Delta V_{th,sm}$ denotes the contribution of the small geometry effects (equation 115, Section 10.4)
- ϕ_{Spg} denotes the contribution of the poly-Si gate depletion effects (equation 73, Section 6)

The modeled ΔV_{th} is incorporated in the ϕ_S iteration as can be seen in Eq. (10), and can be viewed as consisting of two main effects or components:

(I) Short-channel effect: $\Delta V_{th,SC}$

(II) Reverse-short-channel effect: $\Delta V_{th,R}$ and $\Delta V_{th,P}$

The separation into these two components ($\Delta V_{th} = \Delta V_{th,SC} + \Delta V_{th,R}$ (or $\Delta V_{th,P}$)) is schematically shown in Fig. 6.

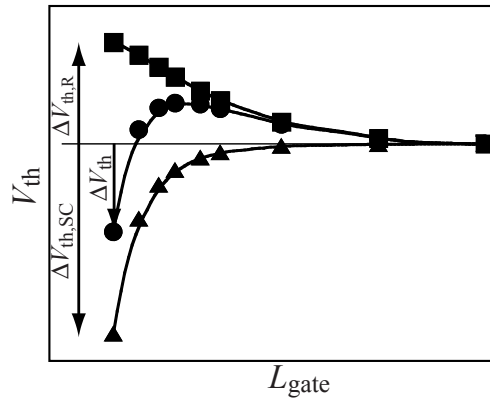


Fig. 6: Schematic plot of the separation of V_{th} into the contributions of the short-channel and the reverse-short-channel effect.

The short-channel effect must be known prior to the potential calculation, and the internal node potential must be explicitly determined. Therefore, V'_{bs} in the following equations is equal to zero for the floating body and to V_{bcs} for the body tie.

4.1 (I) Short-Channel Effects

By applying the Gauss's law, $\Delta V_{th,SC}$ can be written as a function of the lateral electric field E_y . A parabolic potential distribution along the channel is approximated, which results in a position-independent gradient of the lateral electric field $\frac{dE_y}{dy}$ [4]

$$\Delta V_{th,SC} = \frac{\epsilon_{Si}}{C_{FOX}} W_d \frac{dE_y}{dy} \quad (38)$$

where W_d is the depletion-layer thickness written with the impurity concentration N_{subs} (see Eq. (52)) as

$$W_d = \sqrt{\frac{2\epsilon_{Si}(2\Phi_B - V'_{bs})}{qN_{sub}}} \quad (39)$$

where

$$2\Phi_B = \frac{2 \ln \left(\frac{N_{sub}}{n_i} \right)}{\beta} \quad (40)$$

$\frac{dE_y}{dy}$ is derived with model parameters in the form

$$\frac{dE_y}{dy} = \frac{2(\mathbf{VBI} - 2\Phi_B)}{(L_{gate} - \mathbf{PARL2})^2} \left(\mathbf{SC1} + \mathbf{SC2} \cdot V_{ds} + \mathbf{SC3} \cdot \frac{2\Phi_B - V'_{bs}}{L_{gate}} \right) \quad (41)$$

VBI and **PARL2** represent the built-in potential and the depletion width of the junction vertical to the channel, respectively. The model parameter **SC1** determines the threshold voltage shift for small V_{ds} and V'_{bs} , and is expected to be unity. If measured V_{th} is plotted as a function of V_{ds} , it shows nearly a linear dependence. The gradient is proportional to **SC2**. **SC3** implements a correction of the charge-sheet approximation and is expected to be small. Thus this parameter describes modification of the V_{th} on V'_{bs} for inhomogeneous substrate impurity profile.

In SOI-MOSFET, additional short channel effect is observed because of the electric field via BOX. This effect is modeled as follows [9]:

$$\Delta V_{th,SCR} = \frac{\mathbf{SCR1} \cdot \mathbf{TFOX} (E_g + 2\Phi_B - \mathbf{SCR3} + \mathbf{SCR2} \cdot V_{ds})}{\frac{L_{gate}}{2} + \mathbf{PARL1}} \quad (42)$$

4.2 (II) Reverse-Short-Channel Effects

The reverse-short-channel effect is categorized into resulting from two physical MOSFET properties:

- (i) **Impurity concentration inhomogeneity in the direction vertical to the channel (vertical channel inhomogeneity)** (obvious in the retrograded implantation): $\Delta V_{th,R}$
- (ii) **Impurity concentration inhomogeneity in the direction parallel to the channel (lateral channel inhomogeneity)** (obvious in the pocket implantation): $\Delta V_{th,P}$

(i) Impurity concentration inhomogeneity in the direction vertical to the channel (Retrograded Implantation)

The model part described in section 4.2.(i) is not included in HiSIM_SOI explicitly, but the above model parameters **SC3** and **SCP3** can be used to describe the effect successfully, if the inhomogeneity is not extremely large.

(ii) Impurity concentration inhomogeneity in the lateral direction parallel to the channel (Pocket Implantation)

The pocket-implant technology causes an inhomogeneity along the channel. Two model parameters (**LP**: length of the pocket extension into the channel; **NSUBP**: peak of the pocket impurity concentration) are introduced. The impurity concentration in the SOI substrate is distinguished with **NSUBS** from **NSUBP** [1,8]. The resulting model equations for the V_{th} shift due to the pocket implant are:

$$\Delta V_{th,P} = (V_{th,R} - V_{th0}) \frac{\epsilon_{Si}}{C_{FOX}} W_d \frac{dE_{y,P}}{dy} \quad (43)$$

$$V_{th,R} = V_{FB} + 2\Phi_B + \frac{Q_{B0}}{C_{FOX}} + P_{tovr} \quad (44)$$

$$Q_{B0} = \sqrt{2q \cdot N_{sub} \cdot \epsilon_{Si} \cdot (2\Phi_B - V'_{bs})} \quad (45)$$

$$V_{th0} = V_{FB} + 2\Phi_{BC} + \frac{\sqrt{2q \cdot N_{subs} \cdot \epsilon_{Si} \cdot (2\Phi_{BC} - V'_{bs})}}{C_{FOX}} \quad (46)$$

$$\frac{dE_{y,P}}{dy} = \frac{2(V_{BI} - 2\Phi_B)}{LP^2} \left(\mathbf{SCP1} + \mathbf{SCP2} \cdot V_{ds} + \mathbf{SCP3} \cdot \frac{2\Phi_B - V'_{bs}}{LP} \right) \quad (47)$$

$$P_{tovr} = \begin{cases} \frac{1}{\beta} \ln \left(\frac{N_{subb0}}{N_{subs}} \right) & (L_{gate} \leq 2 \cdot LP) \\ 0 & (L_{gate} > 2 \cdot LP) \end{cases} \quad (48)$$

$$N_{subb0} = 2 \cdot N_{subps} - \frac{(N_{subps} - N_{subs}) \cdot L_{gate}}{LP} - N_{subs} \quad (49)$$

$V_{th,R}$ and V_{th0} , defined in Eqs. (44) and (46), are the threshold voltages for the cases with and without pocket-implant, respectively. The parameters **SCP1** – **SCP3** describe the short-channel effect caused by the potential minimum at the higher impurity concentration of the pocket. $2\Phi_{BC}$ is the potential giving threshold condition with **NSUBS** and $2\Phi_B$ is the equivalent potential with N_{subs} as follows:

$$\Phi_{BC} = \frac{2}{\beta} \ln \left(\frac{N_{subs}}{n_i} \right) \quad (50)$$

$$\Phi_B = \frac{2}{\beta} \ln \left(\frac{N_{sub}}{n_i} \right) \quad (51)$$

$$N_{sub} = \begin{cases} \frac{N_{subs}(L_{gate} - LP) + N_{subps} \cdot LP}{L_{gate}} & (L_{gate} > LP) \\ N_{subps} + \frac{(N_{subps} - N_{subs}) \cdot (LP - L_{gate})}{LP} & (L_{gate} \leq LP) \end{cases} \quad (52)$$

As defined in Eq. (52), N_{subs} (see (101)) replaced to the averaged impurity concentration in the channel and N_{subps} is introduced, beginning from channel lengths where pockets at source and drain start to overlap.

Usually strong pocket implantation induces a vertical impurity distribution at the same time. For fitting the measured results in such cases it is recommended to use the parameter **SCP3**.

4.3 Flat-Band Voltage Shift due to Mechanical Stress

It is known that the stress induced on the gate changes the work function value, which causes the flat-band shift. This effect is introduced as

$$V_{fb} = f(V_{fb1}, V_{fb2}, V_{fb3}) \quad (53)$$

where f means a function. In this function, V_{fb1} is smoothly limited not to exceed V_{fb2} , and the resultant value is also smoothly limited not to exceed V_{fb3} . Those intermediate values V_{fb1} , V_{fb2} and V_{fb3} are introduced for the modeling of the gate-length dependence in the following expressions

$$V_{fb1} = \mathbf{VFBC} \cdot \left(1 + \frac{\mathbf{VFBC1}}{(L_{gate} \cdot 10^6)^{\mathbf{VFBC1P}}} \right) \quad (54)$$

$$V_{fb2} = \mathbf{VFBC} \cdot \left(1 + \frac{\mathbf{VFBC2}}{(L_{gate} \cdot 10^6)^{\mathbf{VFBC2P}}} \right) \quad (55)$$

$$V_{fb3} = \mathbf{VFBC} + \mathbf{VFBHAMP} \cdot (L_{gate} \cdot 10^6) \quad (56)$$

Table 3: HiSIM_SOI model parameters introduced in Section 4 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

VBI	built-in potential
PARL1	SOI SCE parameter
PARL2	depletion width of channel/contact junction
SC1	magnitude of short-channel effect
SC2	V_{ds} -dependence of short-channel effect
*SC3	V_{es} -dependence of short-channel effect
SCR1	parameter for SCE via BOX
SCR2	parameter for SCE via BOX
SCR3	parameter for SCE via BOX
SCP1	magnitude of short-channel effect due to pocket
SCP2	V_{ds} -dependence of SCE due to pocke
*SCP3	V_{es} -dependence of SCE due to pocke
LP	pocket penetration length
NSUBB	substrate impurity concentration
NSUBP	max pocket concentration
VFBC	flat-band voltage
VFBC1	channel length dependence of flat-band voltage
VFBC1P	channel length dependence of flat-band voltage
VFBC2	channel length dependence of flat-band voltage
VFBC2P	channel length dependence of flat-band voltage
VFBHAMP	channel length dependence of flat-band voltage

5 Punchthrough

In idealized MOSFETs, the gate voltage controls carrier concentration at the semiconductor substrate surface. However, this electrostatical control begins to be compromised due to the proximity of source and drain contacts, when MOSFET geometry scales down. Thus, in addition to carriers confined at the surface but also those deep from the surface tend to flow.

Thus the drain current consists of MOSFET intrinsic drain current, $I_{ds,intrinsic}$, and additional leakage types of current, which is punchthrough current $I_{ds,punch}$ and pinch-off $I_{ds,pinchoff}$:

$$I_{ds} = I_{ds,intrinsic} + I_{ds,punch} + I_{ds,pinchoff} \quad (57)$$

The origin of the punchthrough effect is the bipolar effect through source, substrate, and drain. The flag **COPT** is introduced to activate the punchthrough currents:

- **COPT** = 0: without punchthrough effects
- **COPT** = 1: with punchthrough effects

Two mechanisms for the punchthrough effect are considered:

$$I_{ds,punch} = I_{surfacePT} + I_{deepPT} \quad (58)$$

5.1 Shallow Punchthrough Effect

The effect is described by a power function of the potential difference instead of the exponential function as

$$POTEN = \left(1.1 - \left(\phi_{s0,SOI} + \frac{V_{ds} - (\phi_{sL,SOI} - \phi_{s0,SOI})}{2} \right) \right)^{PTP} \quad (59)$$

$$I_{shallowPT} = \frac{W_{eff} \cdot \mathbf{NF}}{L_{eff}} \frac{\mu}{\beta} \cdot (\phi_{sL,SOI} - \phi_{s0,SOI}) \left\{ C_{FOX} \cdot \beta \frac{\mathbf{PTL}}{(L_{gate} \cdot 10^6)^{PTLP}} \cdot POTEN \cdot \left(1 + \mathbf{PT2} \cdot V_{ds} + \frac{\mathbf{PT4} \cdot (\phi_{s0,SOI} - V_{bs})}{(L_{gate} \cdot 10^6)^{PT4P}} \right) \right\} \quad (60)$$

where model parameters **PTL**, **PTLP**, **PT2**, **PT4**, and **PT4P** are introduced. Note surface punchthrough effects can be also turned off by setting **PTL**=0.

5.2 Deep Punchthrough Effect:new

Apart from the shallow punchthrough current, a direct passage of carriers between source and drain can be formed deep beneath the surface, at the depth comparable to the source and drain junction depth at most.

The deep punchthrough current, I_{deepPT} , is written as:

$$I_{PT,deep} = J_{PT,deep} \cdot W_{eff} \cdot \mathbf{NF} \cdot (1 - \exp(-\beta V_{ds})) \quad (61)$$

A prefactor $J_{PT,deep}$ is expressed as:

$$J_{PT,deep} = \frac{2}{\beta L_{eff}} \cdot Q_{n0,PT} \cdot \mathbf{MUPT} \cdot \exp(-\beta \phi_m) \quad (62)$$

MUPT represents an effective mobility for deep punchthrough current, and $Q_{n0,PT}$ represents an effective charge density per unit area for deep punchthrough current.

$$Q_{n0,PT} = \sqrt{\frac{2 \cdot q \cdot \mathbf{NJUNC} \cdot \varepsilon_{si}}{\beta}} \sqrt{\beta \phi_{m,gate}} \quad (63)$$

where **NJUNC** is dopant concentration at source and drain junction part.

ϕ_m represents an effective potential barrier for carriers to summount from the source to flow into the channel and is regarded as an interplay of electrostatic effects by V_{gs} and V_{ds} , which will be modeled separately:

$$\phi_m = \phi_{m,gate} + \phi_{m,SD} \quad (64)$$

where and $\phi_{m,SD}$ is that by V_{ds} .

$$\phi_{m,gate} = (\phi_{s,deepPT} - \phi_{m,SD}) * wfactor \quad (65)$$

$$wfactor = \begin{cases} \left(1 - \frac{\mathbf{XJPT}}{W_{depl,PT}}\right)^2 & (W_{depl,PT} \geq \mathbf{XJPT}) \\ 0 & (W_{depl,PT} < \mathbf{XJPT}) \end{cases} \quad (66)$$

$$W_{depl,PT} = \begin{cases} \frac{Q_{bu,PT}}{q \cdot N_{subs}} & (\phi_{s,deepPT} \geq \phi_{m,SD}) \\ 0 & (\phi_{s,deepPT} < \phi_{m,SD}) \end{cases} \quad (67)$$

$$Q_{bu,PT} = \text{const0} \cdot \sqrt{\exp(-\beta \cdot (\phi_{s,deepPT} - \phi_{m,SD})) - 1 + \beta \cdot (\phi_{s,deepPT} - \phi_{m,SD})} \quad (68)$$

The V_{ds} controlled part of ϕ_m is expressed as:

$$\phi_{m,SD} = -\frac{1}{4} \frac{(E_{cri} \cdot L_{eff})^2}{(E_{cri} \cdot L_{eff}) + V_{ds}} \quad (69)$$

where E_{cri} is the critical electric field strength at the junction:

$$E_{cri} = \sqrt{\frac{2q \cdot (V_{bi} - V_{bs})}{\varepsilon_{si}} \frac{N_{sub} \cdot \mathbf{NJUNC}}{N_{sub} + \mathbf{NJUNC}}} \quad (70)$$

where V_{bi} is built-in potential.

$\phi_{s,deepPT}$ is the potential for the deep region where the deep punchthrough occurs, calculated with use of the Poisson equation with a different flat-band voltage:

$$\mathbf{VFBC} + \mathbf{VFBPT} \quad (71)$$

Additionally, **PSLIMPT** is introduced to limit the $\phi_{s,deepPT}$, namely the maximum value.

5.3 Channel Conductance

The high field under the saturation condition causes the pinch-off region and the current flow diverts away from the surface. This effect is considered as the lateral-field-induced charge for the capacitance (see section 14). The simplified formulation is applied to consider the effect as

$$I_{ds,pinchoff} = \frac{W_{eff} \cdot \mathbf{NF}}{L_{eff}} \frac{\mu}{\beta} \cdot (\phi_{sL,SOI} - \phi_{s0,SOI}) \cdot \text{COND} \quad (72)$$

$$\text{COND} = C_{FOX} \cdot \beta \frac{\mathbf{GDL}}{(L_{gate} \cdot 10^6 + \mathbf{GDLD} \cdot 10^6)^{\mathbf{GDLP}}} \cdot V_{ds}$$

5.4 Deactivation of the punchthrough models

The punchthrough models above can be turned off by setting a model parameter adequately:

- **PTL** =0 : no shallow punchthrough (5.1)
- **MUPT**=0 : no deep punchthrough (5.2)
- **GDL**=0 : no channel conductance (5.3)

Table 4: HiSIM_SOI model parameters introduced in Section 5 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

PTL	strength of front/back current
PTLP	L_{gate} dependence of front/back current
PTP	strength of front/back current
PT2	V_{ds} dependence of front/back current
PT4	V_{es} dependence of front/back current
PT4P	V_{es} dependence of front/back current
NJUNC	source/drain diffusion dopant concentration
XJPT	effective junction depth for punchthrough current calculation
MUPT	effective mobility for punchthrough current
PSLIMPT	potential limiter for punchthrough current calculation
VFBPT	flatband voltage shifter for punchthrough current
*GDL	strength of high-field effect in front current
*GDLP	L_{gate} dependent high-field effect in front current
*GDLD	L_{gate} dependent high-field effect in front current

6 Depletion Effect of the Gate Poly-Si

To simplify the calculation procedure for the circuit-simulation application, the poly-depletion effect, the potential drop within the poly-Si ϕ_{SpG} , is approximated as a function of V_{gs} and V_{ds} as shown in Eq. (73), and is included in the ΔV_{th} calculation as a potential drop of V_{gs} [10]

$$\phi_{\text{SpG}} = \text{PGD1} \left(1 + \frac{1}{L_{\text{gate}} \times 10^6} \right)^{\text{PGD4}} \exp \left(\frac{V_{\text{gs}} - \text{PGD2}}{[1V]} \right) \quad (73)$$

In Eq. (73), **PGD1** describes the strength of the poly-depletion, **PGD2** represents the threshold voltage of the poly-depletion, and **PGD4** is introduced to describe the channel length dependence. The function of Eq. (73) is smoothed so that ϕ_{SpG} does not exceed ϕ_{s0} for very large V_{gs} . $[1V]$ in the exponential term is introduced just to delete dimension.

Table 5: HiSIM_SOI model parameters introduced in Section 6 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

PGD1	strength of poly-depletion effect
PGD2	threshold voltage of poly-depletion effect
*PGD4	L_{gate} -dependence of poly-depletion effect

7 Quantum-Mechanical Effects

The main quantum-mechanical phenomenon, which has to be included into a MOSFET model for circuit simulation, is the repulsion of the channel's carrier-density peak into the substrate away from the surface. This can be described phenomenologically by an increased effective oxide thickness ΔT_{FOX} .

After some simplifications final equations implemented into HiSIM for the reproduction of quantum mechanical effects are:

$$T_{\text{FOX}} = \text{TFOX} + \Delta T_{\text{FOX}} \quad (74)$$

$$\Delta T_{\text{ox}} = \frac{\text{QME1}}{V_{\text{gs}} - V'_{\text{bs}} - V_{\text{th}}(T_{\text{FOX}} = \text{TFOX}) + \text{QME2}} + \text{QME3} \quad (75)$$

where **QME1**, **QME2**, and **QME3** are the quantum-effect model parameters. The node potential V'_{bs} is set to zero for the floating body and to V_{bcs} for the body tie.

Table 6: HiSIM_SOI model parameters introduced in Section 7 of this manual.

QME1	V_{gs} -dependence of quantum mechanical effect
QME2	V_{gs} -dependence of quantum mechanical effect
QME3	minimum T_{FOX} modification

8 Mobility Model

The low-field mobility is described with the following expressions and includes the three independent mechanisms of Coulomb, phonon and surface-roughness scattering [11]:

$$\frac{1}{\mu_0} = \frac{1}{\mu_{CB}} + \frac{1}{\mu_{PH}} + \frac{1}{\mu_{SR}} \quad (76)$$

$$\mu_{CB}(\text{Coulomb}) = \mathbf{MUECB0} + \mathbf{MUECB1} \cdot \frac{Q_i}{q \times 10^{11}} \quad (77)$$

$$\mu_{PH}(\text{phonon}) = \frac{M_{uephonon}}{E_{eff}^{\mathbf{MUEPH0}}} \quad (78)$$

$$\mu_{SR}(\text{surface roughness}) = \frac{\mathbf{MUESR1}}{E_{eff}^{M_{uesurface}}} \quad (79)$$

where $\mu_{PH}(\text{phonon})$ is temperature dependent as modeled in section 12. Here E_{eff} is the effective field normal to the surface.

$$E_{eff} = \frac{1}{\epsilon_{Si}} (\mathbf{NDEP} \cdot Q_{b,SOI} + \mathbf{NINV} \cdot Q_i) \left(\frac{1}{1 + (\phi_{sL,SOI} - \phi_{s0,SOI})\mathbf{NINVD}} \right) \quad (80)$$

$$Q_{b,SOI} = \frac{Q_{b,SOI}(L) + Q_{b,SOI}(0)}{2} \quad (81)$$

$$Q_i = \frac{Q_i(L) + Q_i(0)}{2} \quad (82)$$

The mobility universality preserves following conditions [12, 13]

$$\mathbf{MUEPH0} \simeq 0.3 \quad (83)$$

$$M_{uesurface} = 2.0 \quad (84)$$

$$\mathbf{NDEP} = 1.0 \quad (85)$$

$$\mathbf{NINV} = 0.5 \quad (86)$$

independent of technology variations. However, these parameters can be used for fitting purposes [14], if it is necessary. Due to the carrier flow at increasing distance from the surface with reducing L_{gate} , the electric field experienced by the carriers is different from the field in the long L_{gate} case. This results in a modification of $M_{uephonon}$, which is modeled as

$$M_{uephonon} = \mathbf{MUEPH1} \times \left(1 + \frac{\mathbf{MUEPHL}}{(L_{gate} \times 10^6)^{\mathbf{MUEPLP}}} \right) \quad (87)$$

The surface-roughness coefficient $M_{uesurface}$ is modeled to have a similar channel length dependence written as

$$M_{uesurface} = \mathbf{MUESR0} \times \left(1 + \frac{\mathbf{MUESRL}}{(L_{gate} \times 10^6)^{\mathbf{MUESLP}}} \right) \quad (88)$$

because the same physical reasoning as for the $M_{uephonon}$ dependence on L_{gate} applies.

The high-field mobility is modeled as [15]

$$\mu = \frac{\mu_0}{\left(1 + \left(\frac{\mu_0 E_y}{V_{max}} \right)^{\mathbf{BB}} \right)^{\frac{1}{\mathbf{BB}}}} \quad (89)$$

$$E_y = \sqrt{\left(\frac{I_{dd}}{\beta \cdot Q_i(0) \cdot (L_{eff} \cdot 10^6)} \right)^2 + \left(\frac{0.2 \cdot V_{max}}{\mu_0} \right)^2} \quad (90)$$

where the maximum velocity V_{max} is temperature dependent modeled, as described in section 12. $\mathbf{BB}$ is usually fixed to 2 for electrons and 1 for holes. E_y is derived from calculated ϕ_s values. The maximum velocity V_{max} should be the maximum electron-saturation velocity ($\simeq 1 \times 10^7 \text{ cm/s}$), which is however exceeded at reduced L_{gate} . This phenomenon, called velocity overshoot, is included in the mobility model in the following manner

$$V_{max} = \mathbf{VMAX} \cdot \left(1 + \frac{\mathbf{VOVER}}{(L_{gate} \times 10^6)^{\mathbf{VOVERP}}} \right) \quad (91)$$

Table 7: HiSIM_SOI model parameters introduced in Section 8 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

MUECB0	Coulomb scattering for current
MUECB1	Coulomb scattering for current
MUEPH0	phonon scattering for current
MUEPH1	phonon scattering for current
*MUEPHL	length dependence of phonon mobility reduction for current
*MUEPLP	length dependence of phonon mobility for current
MUESR0	surface-roughness scattering for current
MUESR1	surface-roughness scattering for current
*MUESRL	length dependence of surface roughness mobility reduction for current
*MUESLP	length dependence of surface roughness mobility reduction for current
NDEP	depletion charge contribution on effective-electric field
NINV	inversion charge contribution on effective-electric field
*NINVD	reduced resistance effect for small V_{ds}
BB	high-field-mobility degradation
VMAX	maximum saturation velocity
VOVER	velocity overshoot effect
VOVERP	L_{eff} dependence of velocity overshoot

9 Channel-Length Modulation

The strength of the channel-length modulation is dependent on the junction profile between the channel and the drain contact. This dependence is modeled with the parameter **CLM1** as

$$\phi_{s,\text{SOI}}(\Delta L) = (1 - \mathbf{CLM1}) \cdot \phi_{sL,\text{SOI}} + \mathbf{CLM1} \cdot (\phi_{s0,\text{SOI}} + V_{ds}) \quad (92)$$

where **CLM1** can be interpreted to represent the abruptness of the junction and must be in the range $0 \leq \mathbf{CLM1} \leq 1$. Here **CLM1** = 1 means that the contact profile is abrupt and the complete potential increase occurs within the pinch-off region, whereas **CLM1** = 0 corresponds to the opposite condition and there is no steep potential increase in the pinch-off region but the most potential increase occurs within the overlap region.

The length of the pinch-off region ΔL is derived as

$$\Delta L = \frac{1}{2} \left[-\frac{1}{L_{\text{eff}}} \left(2 \frac{I_{dd}}{\beta Q_i} z + 2 \frac{q N_{\text{subs}}}{\epsilon_{\text{Si}}} (\phi_{s,\text{SOI}}(\Delta L) - \phi_{sL,\text{SOI}}) z^2 + E_0 z^2 \right) \right. \\ \left. + \sqrt{\frac{1}{L_{\text{eff}}^2} \left(2 \frac{I_{dd}}{\beta Q_i} z + 2 \frac{q N_{\text{subs}}}{\epsilon_{\text{Si}}} (\phi_{s,\text{SOI}}(\Delta L) - \phi_{sL,\text{SOI}}) z^2 + E_0 z^2 \right)^2 + 4 \left(2 \frac{q N_{\text{subs}}}{\epsilon_{\text{Si}}} (\phi_{s,\text{SOI}}(\Delta L) - \phi_{sL,\text{SOI}}) z^2 + E_0 z^2 \right)} \right] \quad (93)$$

where E_0 is fixed to 10^5 and

$$z = \frac{\epsilon_{\text{Si}} W_d}{C_{\text{LM2}} \cdot Q_b + \mathbf{CLM3} \cdot Q_i} \quad (94)$$

where

$$C_{\text{LM2}} = \begin{cases} \frac{5.0 \cdot 10^9}{\text{TSOI} \cdot \text{NSUBS}} & (\text{default}) \\ \mathbf{CLM2} & (\text{if } \mathbf{CLM2} \text{ is given in the model card.}) \end{cases} \quad (95)$$

Two model parameters **CLM2** and **CLM3** are introduced to consider the uncertainty of Q_i in the pinch-off region and to counterbalance the two contributions from Q_b ($= q N_{\text{subs}} W_d$) and Q_i . It has to be notified that ΔL is equal to zero, when **CLM1** = 0.

Though ΔL is determined mostly by $\phi_{s,\text{SOI}}(\Delta L)$, the combination between **CLM2** and **CLM3** influences the lateral-field-induced capacitance C_{Q_y} described in section 14.

For the pocket implantation the lateral field increase is observed even for long-channel transistors. The effect is modeled by the threshold voltage shift. However, the degradation of the channel conductance under the saturation condition is much stronger than that described by the threshold voltage shift alone. This additional pocket effect is modeled as

$$\Delta L \leftarrow \Delta L (1 + \mathbf{CLM6} \cdot (L_{\text{gate}} \times 10^6)^{\mathbf{CLM5}}) \quad (96)$$

It can happen that $L_{\text{eff}} - \Delta L$ becomes negative, if extracted **CLM5** and **CLM6** values are out of acceptable ranges. In this case HiSIM gives “warning” and fixes $L_{\text{eff}} - \Delta L$ to 1 nm.

Table 8: HiSIM_SOI model parameters introduced in Section 9 of this manual.

CLM1	abruptness coefficient of channel/contact junction
CLM2	coefficient for Q_b contribution
CLM3	coefficient for Q_i contribution
CLM5	CLM parameter
CLM6	CLM parameter

10 Narrow-Channel Effects

10.1 Threshold Voltage Modification

The V_{th} reduction for reduced channel width (W_{gate}) is modeled under inclusion of the edge-fringing capacitances C_{ef} at the edge of the trench [16] as

$$\Delta V_{th,W} = \left(\frac{1}{C_{FOX}} - \frac{1}{C_{FOX} + 2C_{ef}/(L_{eff}W_{eff})} \right) qN_{subc}W_d + \frac{\mathbf{WVTH0}}{W_{gate} \times 10^6} \quad (97)$$

where **WVTH0** is the parameter for including the basic width dependence and

$$C_{ef} = \frac{2\epsilon_{ox}}{\pi} L_{eff} \ln \left(\frac{2 \cdot T_{oxiso}}{T_{FOX}} \right) \quad (98)$$

$$= \frac{\mathbf{WFC}}{2} L_{eff} \quad (99)$$

where T_{oxiso} is an oxide effective thickness at the device isolation edge, which value would not be well determined. Here, **WFC** is the model parameter for including the edge-fringing-capacitance effects.

Enhancement of the V_{th} reduction with decreasing W_{gate} is often measured when L_{gate} is also small. This effect becomes obvious for the pocket technology and is modeled by a modification of the pocket impurity concentration as

$$N_{subpp} = \mathbf{NSUBP} \times \left(1 + \frac{\mathbf{NSUBP0}}{(W_{gate} \times 10^6)^{\mathbf{NSUBWP}}} \right) \quad (100)$$

The device-size dependence of the substrate impurity concentration N_{subs} is also considered as

$$N_{subs} = \min \left(\mathbf{NSUBCMAX}, \mathbf{NSUBS} \cdot \left(1 + \frac{\mathbf{NSUBCW}}{(W_{gate} \cdot 10^6)^{\mathbf{NSUBCWP}}} \right) \cdot \left(1 + \frac{\mathbf{NSUBCL}}{(L_{gate} \cdot 10^6)^{\mathbf{NSUBCLP}}} \right) \right) \quad (101)$$

where **NSUBCMAX** determines the maximum **NSUBS** value.

10.2 Mobility Change

It is known that the trench isolation induces mechanical stress in the channel, which results in a degradation of the mobility [17], and causes a reduction of the saturation current with reduced W_{gate} . This effect is modeled by a decreasing phonon mobility with two model parameters **MUEPHW** and **MUEPWP** as

$$M_{uephonon} \leftarrow M_{uephonon} \times \left(1 + \frac{\mathbf{MUEPHW}}{(W_{gate} \times 10^6)^{\mathbf{MUEPWP}}} \right) \quad (102)$$

However, the I_{ds} - W_{gate} characteristic does not show monotonous decrease but starts to increase for narrower W_{gate} . This effect is modeled as a change of the surface-roughness contribution caused by a carrier flow in increasing distance from the surface as

$$M_{uesurface} \leftarrow M_{uesurface} \times \left(1 + \frac{\mathbf{MUESRW}}{(W_{gate} \times 10^6)^{\mathbf{MUESWP}}} \right) \quad (103)$$

10.3 Transistor Leakage due to Shallow Trench Isolation (STI): Hump in I_{ds}

The shallow trench isolation induces also an undesired hump in the subthreshold region of the I_{ds} - V_{gs} characteristics. A MOSFET leakage current occurs at the STI edges, which is smaller than the main MOSFET current, and only important for modeling of the subthreshold characteristics of the MOSFET. The surface potential of the leakage regions at the trench edges can be derived analytically as in [18]

$$\phi_{s,STI} = V'_{gs,STI} + \frac{\epsilon_{Si}Q_{N,STI}}{C_{FOX}'} \left[1 - \sqrt{1 + \frac{2C_{FOX}'^2}{\epsilon_{Si}Q_{N,STI}} \left(\beta \cdot (V'_{gs,STI} - V'_{bs}) - 1 \right)} \right] \quad (104)$$

where

$$Q_{N,STI} = q \cdot \mathbf{NSTI} \quad (105)$$

$$V'_{gs,STI} = V_{gs} - \mathbf{VFBC} + V_{thSTI} + \Delta V_{th,SCSTI} \quad (106)$$

where

$$V_{thSTI} = \mathbf{VTHSTI} - \mathbf{VDSTI} \cdot V_{ds} \quad (107)$$

and

$$\Delta V_{th,SCSTI} = \frac{\epsilon_{Si}}{C_{FOX}} W_{d,STI} \frac{dE_y}{dy} \quad (108)$$

The threshold voltage for the STI effect $\mathbf{VTHSTI}$ includes features of STI such as $\mathbf{NSTI}$, which are different from the substrate. The depletion-layer thickness $W_{d,STI}$ is written as

$$W_{d,STI} = \sqrt{\frac{2\epsilon_{Si}(2\Phi_{B,STI} - V'_{bs})}{q \cdot \mathbf{NSTI}}} \quad (109)$$

where V'_{bs} is set to zero for the floating body and to V_{bcs} for the body tie. $\frac{dE_y}{dy}$ is described with model parameters in the same form as in section 4.1 on short-channel effects

$$\frac{dE_y}{dy} = \frac{2(\mathbf{VBI} - 2\Phi_{B,STI})}{(L_{gate,sm} - \mathbf{PARL2})^2} (\mathbf{SCSTI1} + \mathbf{SCSTI2} \cdot V_{ds}), \quad (110)$$

where

$$L_{gate,sm} = L_{gate} + \frac{\mathbf{WL1}}{\mathbf{wl} \mathbf{WL1P}} \quad (111)$$

$$\mathbf{wl} = (W_{gate} \cdot 10^6) \cdot (L_{gate} \cdot 10^6) \quad (112)$$

The modeling of the transistor leakage for STI technologies is based on the idea that the current in the subthreshold region is governed only by the diffusion term. The carrier concentration $Q_{i,STI}$ is calculated analytically by the Poisson equation with the substrate-impurity concentration $\mathbf{NSTI}$ different from $\mathbf{NSUBS}$ and $\mathbf{NSUBP}$. The final leakage current equation is written as

$$I_{ds,STI} = 2 \frac{\mathbf{WSTI}}{L_{eff} - \Delta L} \mu \frac{Q_{i,STI}}{\beta} [1 - \exp(-\beta V_{ds})] \quad (113)$$

where W_{STI} determines the width of the high-field region. The gate length dependence of W_{STI} is included as

$$W_{STI} = \mathbf{WSTI} \left(1 + \frac{\mathbf{WSTIL}}{(L_{gate,sm} \cdot 10^6) \mathbf{WSTILP}} \right) \left(1 + \frac{\mathbf{WSTIW}}{(W_{gate} \cdot 10^6) \mathbf{WSTIWP}} \right) \quad (114)$$

To consider the current caused by the STI effect, the flag $\mathbf{COISTI}$ must be set to 1, where the default is set to zero.

10.4 Small Geometry

Small size devices do not show the same scaling characteristic as long-channel or wide-channel devices, but rather deviate significantly. The small geometry effects are modeled first as the threshold voltage shift, which is portion of the threshold voltage shift (equation (37), Section 4).

$$\Delta V_{th,sm} = \frac{\mathbf{WL2}}{\mathbf{wl} \mathbf{WL2P}} \quad (115)$$

$$\mathbf{wl} = (W_{gate} \times 10^6) \times (L_{gate} \times 10^6) \quad (116)$$

The mobility modification due to the small device geometry is also modeled in the phonon scattering as

$$M_{uephonon} \leftarrow M_{uephonon} \times \left(1 + \frac{\mathbf{MUEPHS}}{\mathbf{wl} \mathbf{MUEPSP}} \right) \quad (117)$$

$$V_{max} \leftarrow V_{max} \cdot \left(1 + \frac{\mathbf{VOVERS}}{\mathbf{wl} \mathbf{VOVERS}} \right) \quad (118)$$

Table 9: HiSIM_SOI model parameters introduced in Section 10 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

* MUEPHW	phonon-related mobility reduction
* MUEPWP	phonon-related mobility reduction
* MUEPHS	mobility change due to small size
* MUEPSP	mobility change due to small size
* MUESRW	surface roughness-related mobility change
* MUESWP	surface roughness-related mobility change
* NSUBCL	modification of substrate concentration for short gate length
* NSUBCLP	modification of substrate concentration for short gate length
* NSUBCW	modification of substrate concentration for narrow width
* NSUBCWP	modification of substrate concentration for narrow width
* NSUBCMAX	upper limit of substrate concentration after device-size scaling
WFC	threshold voltage change due to cap change
* WVTH0	threshold voltage shift
* NSUBP0	modification of pocket concentration for narrow W
* NSUBWP	modification of pocket concentration for narrow W
WL2	threshold voltage shift due to small-size effect
WL2P	threshold voltage shift due to small-size effect
* VOVERS	modification of max velocity due to small size
* VOVERSP	modification of max velocity due to small size
NSTI	substrate-impurity concentration at the STI edge
VTHSTI	threshold voltage shift due to STI
* VDSTI	V_{ds} dependence of threshold voltage shift due to STI
* SCSTI1	the same effect as SC1 but at STI edge
* SCSTI2	the same effect as SC2 but at STI edge
* WL1	threshold voltage shift of STI leakage due to small size effect
* WL1P	threshold voltage shift of STI leakage due to small size effect
* WSTI	width of the high-field region at STI edge
* WSTIL	channel-length dependence of WSTI
* WSTILP	channel-length dependence of WSTI
* WSTIW	channel-width dependence of WSTI
* WSTIWP	channel-width dependence of WSTI

11 Effects of the Source/Drain Diffusion Length for Shallow Trench Isolation (STI) Technologies

The diffusion length, L_{od} between MOSFET gate and STI edge affects the MOSFET characteristics. The influence is observed mainly in V_{th} and in the saturation current, namely in the mobility. The V_{th} change is attributed to a change of the impurity concentration and modeled as

$$N_{substi} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3} \quad (119)$$

where

$$T1 = \frac{1}{1 + NSUBCSTI2} \quad (120)$$

$$T2 = \left(\frac{NSUBCSTI1}{L_{od, half}} \right)^{NSUBCSTI3} \quad (121)$$

$$T3 = \left(\frac{NSUBCSTI1}{L_{od, half, ref}} \right)^{NSUBCSTI3} \quad (122)$$

which is used to modify the channel concentration N_{subs} as

$$N_{subs} = N_{subs} \cdot N_{substi} \quad (123)$$

The V_{th} change is attributed to a change of the pocket impurity concentration and modeled as

$$N_{substi} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3}, \quad (124)$$

where

$$T1 = \frac{1}{1 + NSUBPSTI2} \quad (125)$$

$$T2 = \left(\frac{NSUBPSTI1}{L_{od, half}} \right)^{NSUBPSTI3} \quad (126)$$

$$T3 = \left(\frac{NSUBPSTI1}{L_{od, half, ref}} \right)^{NSUBPSTI3} \quad (127)$$

which is used to modify the pocket concentration N_{subps} as

$$N_{subps} = N_{subpp} \cdot N_{substi} \quad (128)$$

The saturation-current change is attributed to a change of the mobility and modeled as

$$M_{uesti} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3} \quad (129)$$

where

$$T1 = \frac{1}{1 + MUESTI2} \quad (130)$$

$$T2 = \left(\frac{MUESTI1}{L_{od, half}} \right)^{MUESTI3} \quad (131)$$

$$T3 = \left(\frac{MUESTI1}{L_{od, half, eff}} \right)^{MUESTI3} \quad (132)$$

which is used to modify the phonon mobility parameter $M_{uephonon}$ as

$$M_{uephonon} = M_{uephonon} \times M_{uesti} \quad (133)$$

where $L_{od, half}$ and $L_{od, half, eff}$ are determined in the same way as BSIM4.6.0 with model parameters **SAREF** and **SBREF** and instance parameters **SA**, **SB**, and **SD**.

Table 10: HiSIM_SOI model parameters introduced in Section 11 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NSUBPSTI1	pocket concentration modifier
NSUBPSTI2	pocket concentration modifier
NSUBCSTI1	pocket concentration modifier
NSUBCSTI2	pocket concentration modifier
NSUBCSTI3	pocket concentration modifier
MUESTI1	mobility change due to diffusion length
MUESTI2	mobility change due to diffusion length
MUESTI3	mobility change due to diffusion length
LOD	diffusion length between gate and STI edge
SAREF	ref-dist between OD edge to poly of one side
SBREF	ref-dist between OD edge to poly of the other side
SA	length of diffusion between gate and STI
SB	length of diffusion between gate and STI
SD	length of diffusion between gate and gate

12 Temperature Dependence

If **TEMP** is treated as a model parameter, temperature T is determined as

$$T = \mathbf{TEMP} + \mathbf{DTEMP} + 273.15. \quad (134)$$

where **DTEMP** is the temperature increase by self-heating. **DTEMP** can also be an instance parameter **DTEMP**. If **TEMP** is treated as an instance parameter, T is determined as

$$T = \mathbf{TEMP} + 273.15 \quad (135)$$

The nominal temperature **TNOM** is set by

$$TNOM = \mathbf{TNOM} + 273.15. \quad (136)$$

The inverse thermal voltage, respectively for **TNOM** and T , is

$$\beta_{\text{tnom}} = \frac{q}{k_B \cdot TNOM} \quad (137)$$

$$\beta = \frac{q}{k_B T}. \quad (138)$$

The temperature dependence is included automatically through the factor $\beta\phi_s$ that appears in various places in the formulations, where ϕ_s denotes a surface potential ($\phi_{s,\text{bulk}}$, $\phi_{b,\text{SOI}}$, and $\phi_{s,\text{SOI}}$). Additionally the bandgap, the intrinsic carrier concentration, the carrier mobility, and the carrier saturation velocity are also temperature dependent. The temperature dependence of the bandgap determines the temperature dependence of V_{th} [19] and is modeled as

$$E_g = E_{g\text{tnom}} - \mathbf{BGTMP1} \cdot (T - TNOM) - \mathbf{BGTMP2} \cdot (T^2 - TNOM^2) \quad (139)$$

where T is the given temperature, and

$$E_{g\text{tnom}} = \mathbf{EG0} - 90.25 \times 10^{-6} \cdot TNOM - 1.0 \times 10^{-7} \cdot TNOM^2 \quad (140)$$

The temperature dependence of the intrinsic carrier concentration is given by

$$n_i = n_{i0} \cdot T^{\frac{3}{2}} \cdot \exp\left(-\frac{E_g}{2q}\beta\right) \quad (141)$$

The temperature dependence of the mobility and the temperature dependence of the saturation velocity have a major influence on the temperature dependence of the $I_{\text{ds}}-V_{\text{ds}}$ characteristics under the on-current condition. They are modeled as [15]:

$$\mu_{\text{PH}}(\text{phonon}) = \frac{M_{\text{uephonon}}}{(T/\mathbf{TNOM})^{\mathbf{MUETMP}} \times E_{\text{eff}}^{\mathbf{MUEPH0}}} \quad (142)$$

$$V_{\text{max}} = \frac{\mathbf{VMAX}}{1.8 + 0.4(T/\mathbf{TNOM}) + 0.1(T/TNOM)^2 - \mathbf{VTMP} \times (1 - T/\mathbf{TNOM})} \quad (143)$$

where $V_{\text{max}}(T)$, V_{over} and V_{overp} are temperature dependence of **VMAX**, **VOVER** and **VOVERP** in Section 8.

In addition to the temperature dependence of the physical quantities considered, resistances include the temperature dependence, which is modeled with the given temperature to avoid complication in parameter extraction.

$$\mu_{\text{drift0,temp}} = \frac{\mathbf{RDRMUE}}{(T/\mathbf{TNOM})^{\mathbf{RDRMUETMP}}} \quad (144)$$

$$\mu_{\text{source0,temp}} = \frac{\mathbf{RDRMUES}}{(T/\mathbf{TNOM})^{\mathbf{RDRMUETMP}}} \quad (145)$$

$$(146)$$

$$V_{\text{max.drift,temp}} = \frac{\mathbf{RDRVMAX}}{1.8 + 0.4(T/\mathbf{TNOM}) + 0.1(T/\mathbf{TNOM})^2 - \mathbf{RDRV TMP} \cdot (1 - T/\mathbf{TNOM})} \quad (147)$$

$$V_{\text{max.source,temp}} = \frac{\mathbf{RDRVMAXS}}{1.8 + 0.4(T/\mathbf{TNOM}) + 0.1(T/\mathbf{TNOM})^2 - \mathbf{RDRV TMP} \cdot (1 - T/\mathbf{TNOM})} \quad (148)$$

$$R_{\text{drbb,temp}} = \mathbf{RDRBB} + \mathbf{RDRBBTMP}(T - \mathbf{TNOM}) \quad (149)$$

$$R_{\text{srbb,temp}} = \mathbf{RDRBBS} + \mathbf{RDRBBTMP}(T - \mathbf{TNOM}) \quad (150)$$

Table 11: HiSIM_SOI model parameters introduced in Section 12 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

TEMP	device temperature
DTEMP	device temperature change
BGTMP1	temperature dependence of bandgap
BGTMP2	temperature dependence of bandgap
TNOM	temperature selected as a nominal value
EG0	bandgap
MUETMP	temperature-dependence of phonon scattering
*VTMP	temperature-dependence coefficient
RDRBB	high field mobility in drift region (drain side)
RDRBBS	high field mobility in drift region (source side)
RDRMUE	mobility in drift region (drain side)
RDRMUES	mobility in drift region (source side)
RDRVMAX	saturation velocity in drift region (drain side)
RDRVMAXS	saturation velocity in drift region (source side)
RDRVTMP	temperature dependence of resistance
RDRMUETMP	temperature dependence of resistance
RDRBBTMP	temperature dependence of resistance

13 Resistances

Figure 7 shows a schematic feature of the device. The potential drops V_{ssp} and V_{ddp} are calculated by SPICE iteratively.

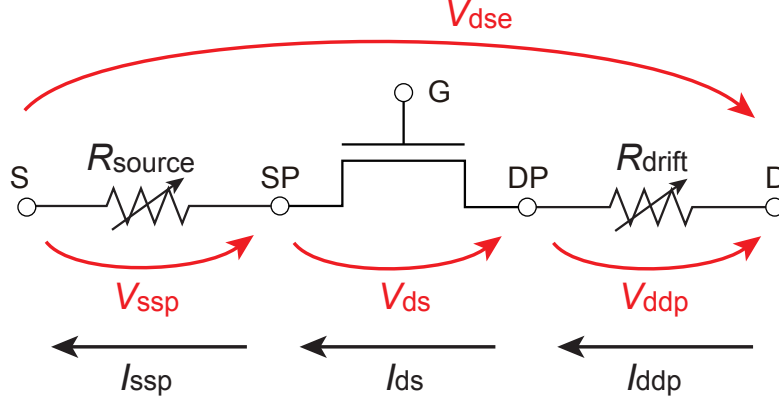


Fig. 7: Model concept

13.1 Drain-side resistance

The drain side resistance R_{drift} is written with the current flowing in the drift region I_{ddp} as

$$R_{drift} = \left(\frac{V_{ddp}}{I_{ddp}} + \mathbf{RSH} \cdot \mathbf{NRD} \right) \quad (151)$$

where $\mathbf{NRD}$ is an instance parameter describing the number of squares of the drain diffusion, and $\mathbf{RSH}$ is its the sheet resistance of the square. The first terms of the right hand side of Eq. (151) considers the resistance in the drift region, and the second term is the that in the diffusion region, which are layout dependent.

$$I_{ddp} = W_{eff} \cdot \mathbf{NF} \cdot X_{ov} \cdot q \cdot \mathbf{NOVER} \cdot \mu_{drift} \frac{V_{ddp}}{\mathbf{LDRIFT}} \quad (152)$$

where

$$\mu_{drift} = \frac{\mu_{drift0}}{\left[1 + \left(\frac{\mu_{drift0}}{V_{max,drift}} \cdot \frac{V_{ddp}}{\mathbf{LDRIFT}} \right) R_{drbb,temp} \right]^{\frac{1}{R_{drbb,temp}}}} \quad (153)$$

$$\mu_{drift0} = \mu_{drift0,temp} \left(1 + \frac{\mathbf{RDRMUEL}}{(L_{gate} \cdot 10^6) \mathbf{RDRMUPL}} \right) \quad (154)$$

$$V_{max,drift} = V_{max,drift,temp} \left(1 + \frac{\mathbf{RDRVMAXL}}{(L_{gate} \cdot 10^6) \mathbf{RDRVMAXLP}} \right) \left(1 + \frac{\mathbf{RDRVMAXW}}{(W_{gate} \cdot 10^6) \mathbf{RDRVMAXWP}} \right) \quad (155)$$

$$X_{ov} = \sqrt{\mathbf{XLD}^2 + \mathbf{RDRDJUNC}^2} \quad (156)$$

The potential value of V_{ddp} , the difference between the internal node DP and the drain node is calculated by circuit simulator.

13.2 Source-side resistance

The source-side resistance R_{source} is written with the current flowing in the drift region I_{ssp} as

$$R_{\text{source}} = \left(\frac{V_{\text{ssp}}}{I_{\text{ssp}}} + \mathbf{RSH} \cdot \mathbf{NRS} \right) \quad (157)$$

where $\mathbf{NRS}$ is an instance parameter describing the number of squares of the source diffusion, and $\mathbf{RSH}$ is its sheet resistance of the square. The first terms of the right hand side of Eq. (157) considers the resistance in the source drift region, and the second term is the that in the source diffusion region, which is layout dependent.

$$I_{\text{ssp}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot X_{\text{ov}} \cdot q \cdot \mathbf{NOVER} \cdot \mu_{\text{source}} \frac{V_{\text{ssp}}}{\mathbf{LDRIFTS}} \quad (158)$$

where

$$\mu_{\text{source}} = \frac{\mu_{\text{source0}}}{\left[1 + \left(\frac{\mu_{\text{source0}}}{V_{\text{max_source}}} \cdot \frac{V_{\text{ssp}}}{\mathbf{LDRIFTS}} \right)^{R_{\text{srbb,temp}}} \right]^{\frac{1}{R_{\text{srbb,temp}}}}} \quad (159)$$

$$\mu_{\text{source0}} = \mu_{\text{source0,temp}} \left(1 + \frac{\mathbf{RDRMUEL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{RDRMUELP}}} \right) \quad (160)$$

$$V_{\text{max_source}} = V_{\text{max_source,temp}} \left(1 + \frac{\mathbf{RDRVMAXL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{RDRVMAXLP}}} \right) \left(1 + \frac{\mathbf{RDRVMAXW}}{(W_{\text{gate}} \cdot 10^6)^{\mathbf{RDRVMAXWP}}} \right) \quad (161)$$

The potential value of V_{ssp} , the difference between the internal node SP and the drain node is calculated by circuit simulator.

13.3 Gate resistance

The gate resistance becomes large as the gate width becomes large, which is the case for RF circuits. The equation for the gate-resistance calculation is taken from BSIM4 [22] description as

$$R_g = \frac{\mathbf{RSHG} \cdot (\mathbf{XGW} + \frac{W_{\text{eff}}}{3 \cdot \mathbf{NGCON}})}{\mathbf{NGCON} \cdot (L_{\text{drawn}} - \mathbf{XGL}) \cdot \mathbf{NF}} \quad (162)$$

where $\mathbf{RSHG}$ is the gate sheet resistance, and others are instance parameters dependent on layout. The flag $\mathbf{CORG}$ is provided for the inclusion of gate resistance. $\mathbf{CORG} = 0, 1$ means "no", "external" gate resistance, respectively.

13.4 Bulk resistance

Internal bulk(body) potential in SOI layer can build up due to substrate current flowing out through the bulk to the body contact. The potential build-up can cause kink in source current and is expressed as:

$$\Delta V_{\text{bs}} = I_{\text{sub}} \cdot R_{\text{bulk}} \quad (163)$$

where I_{sub} represents substrate current and R_{bulk} represents effective bulk resistance between internal bulk beneath the gate and the body contact (body tie). R_{bulk} is expressed as two contributions, one as device-geometry dependent and the other as a constant offset:

$$R_{\text{bulk}} = \mathbf{RBULKW} \cdot W_{\text{eff}} + \mathbf{RBULK0} \quad (164)$$

To facilitate RF modeling, a body resistance network is implemented.

The HiSIM model parameters introduced in section 13 are summarized in Table 12.

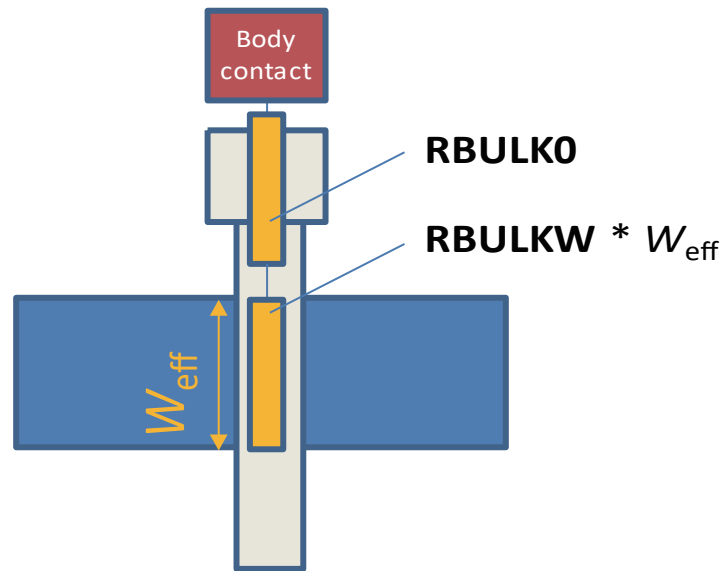


Fig. 8: Bulk resistance

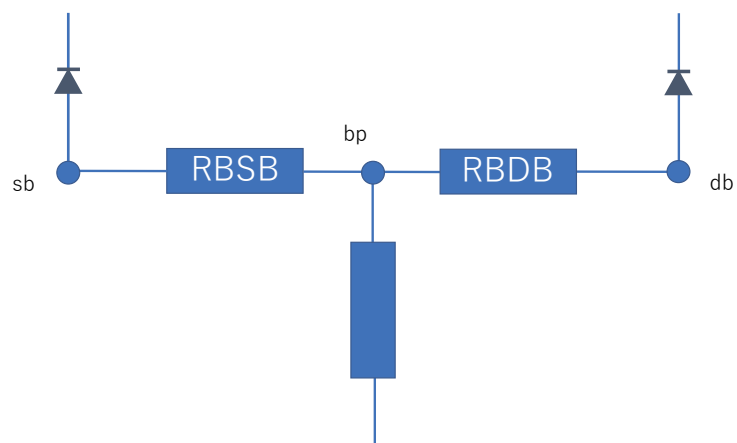


Fig. 9: SOI body resistance network

Table 12: HiSIM model parameters introduced in section 13 of this manual. # indicates instance parameters. * indicates minor parameters.

RSH	sheet resistance of diffusion region (drain side)
RSHG	gate sheet resistance
#NRS	number of source squares
#NRD	number of drain squares
#XGW	distance from the gate contact to the channel edge
#XGL	offset of the gate length
#NF	number of fingers
#NGCON	number of gate contacts
#LDRIFT	length of drift region (drain side)
#LDRIFTS	length of drift region (source side)
XLD	gate overlap length
NOVER	impurity concentration in overlap region
RDRDJUNC	junction depth at channel/drift region (drain side)
RDRVMAXL	saturation velocity L_{gate} dependence
RDRVMAXLP	saturation velocity L_{gate} dependence
RDRVMAXW	saturation velocity W_{gate} dependence
RDRVMAXWP	saturation velocity W_{gate} dependence
RDRMUEL	mobility in drift region L_{gate} dependence
RDRMUELP	mobility in drift region L_{gate} dependence
RBULK0	body resistance (COBCNODE =1)
RBULKW	body resistance coefficient (COBCNODE =1)
#RBSB	body resistance (source side) (CORBNET =1)
#RBDB	body resistance (drain side) (CORBNET =1)

14 Capacitances

14.1 Intrinsic Capacitances

The intrinsic capacitances are derivatives of the node charges determined as

$$C_{jk} = \delta \frac{\partial Q_j}{\partial V_k} \quad (165)$$

$$\delta = -1 \quad \text{for } j \neq k$$

$$\delta = 1 \quad \text{for } j = k$$

HiSIM uses analytical solutions for all independent intrinsic capacitances, derived from the charges as explicit functions of the surface potentials. Therefore, there are no extra model parameters for the intrinsic capacitances except for the length and width reduction parameters, **XLDC** and **XWDC**, which can take a different value from **XLD** and **XWD**, if necessary. In this case, the effective length for the total capacitance calculation, namely, W_{effc} , takes a different value from W_{eff} . Similarly, the effective width for the total capacitance calculation, namely, W_{effc} , takes a different value from W_{eff} .

The lateral electric field along the channel induces a capacitance C_{Q_y} which significantly affects the gate capacitance in saturation [20]. The induced charge associated with C_{Q_y} is described with the surface potential values as

$$Q_y = \epsilon_{\text{Si}} W_{\text{eff}} \cdot \mathbf{NF} \cdot W_d \left(\frac{\phi_{s0} + V_{\text{ds}} - \phi_s(\Delta L)}{\mathbf{XQY}} \right) + \frac{\mathbf{XQY1} \cdot W_{\text{eff}} \times 10^6 \cdot \mathbf{NF}}{(L_{\text{gate}} \times 10^6) \mathbf{XQY2}} V'_{\text{bs}} \quad (166)$$

introducing **XQY**, a parameter determining the maximum field at the channel/drain junction independent of L_{gate} . For **XQY**=0 the charge Q_y is fixed to zero. To compensate the enhanced short-channel effect, determined by the current characteristics, two model parameters **XQY1** and **XQY2** are introduced. To activate Q_y the model parameter **CLM1** must be smaller than unity.

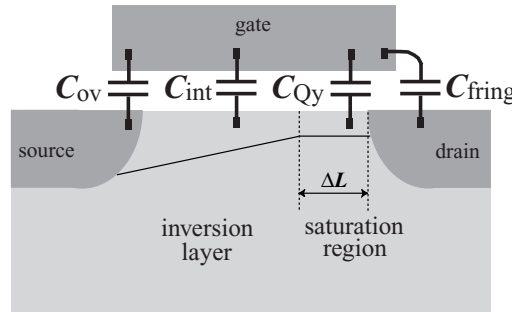


Fig. 10: Modeling of the gate-drain capacitance with C_{Q_y} added to the conventional components.

14.2 Overlap Capacitances

The overlap charge at the drain side Q_{god} is written as

$$\frac{Q_{\text{god}}}{W_{\text{effc}} \cdot \mathbf{NF} \cdot C_{\text{FOX}}} = \int_0^{\text{LOVER}} (V_{\text{gs}} - \phi_s) dy \quad (167)$$

Similarly, at the source side,

$$\frac{Q_{\text{gos}}}{W_{\text{effc}} \cdot \mathbf{NF} \cdot C_{\text{FOX}}} = \int_0^{\text{LOVER}} (V_{\text{gs}} - \phi_s) dy \quad (168)$$

Thus the surface-potential distribution along the gate overlap region determines the charge and the capacitance.

To consider the overlap capacitance, the flag **COADOV** must be selected to 1, which is the default. The overlap capacitance includes three options as summarized in Fig. 11, where the drain side and the source side are treated as identical. If Flag **COOVLP**=0, the overlap capacitances are treated to be constant. If **CGSO** and **CGDO** are defined, these values

are taken. If they are not defined, the values are calculated with the overlap length **LOVER** and oxide capacitance C_{FOX} .

If Flags **COOVLP**=1, the bias dependent overlap capacitances are considered. Here two models are provided: One is the surface-potential-based model and the other describes with a simple V_{gs} dependence. In addition to the bias dependent capacitances, **CGSO** and **CGDO** can be also added, if they are determined.

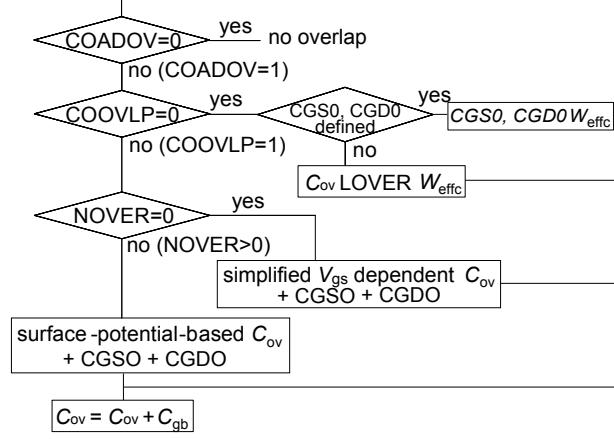


Fig. 11: Model options of the overlap capacitance at the drain side are summarized.

The description is focussed on the drain side. For the source side the same calculation is performed with $V_{ds}=0$.

i) Surface-Potential-Based Model

The surface potential ϕ_s is calculated in the same manner as in the channel region, and only the polarity is inverted from the channel. The final overlap charge equation is written with the calculated ϕ_s

a) under the depletion and the accumulation conditions

$$Q_{over} = W_{effc} \cdot NF \cdot LOVER \left(\sqrt{\frac{2\epsilon_{Si}q \cdot NOVER}{\beta}} \sqrt{\beta(\phi_s + V_{ds}) - 1} \right) \quad (169)$$

b) under the inversion condition

$$Q_{over} = W_{effc} \cdot NF \cdot LOVER \cdot C_{FOX}(V_{gs} - VFBOVER - \phi_s) \quad (170)$$

where **LOVER** is the length of the overlap region of the gate over drain, **NOVER** is the impurity concentration in the drift region, and **VFBOVER** is the flat-band voltage in the overlap region. This model is selected, if **NOVER** is not equal to zero.

Two options are provided to calculate ϕ_s , which is selected by the flag **COQOVSM**:

COQOVSM=0: with an analytical equation excluding inversion charge

COQOVSM=1: with iterative procedure (default)

ii) Simplified Bias-Dependent Model

If **NOVER**=0 and **LOVER** > 0 and the flag **COOVLP** = 1, the overlap charge is modeled as

$$Q_{god} = W_{effc} \cdot NF \cdot C_{FOX} [(V_{gs} - V_{ds})LOVER - OVSLP \cdot (1.2 - (\phi_{sL} - V_{ds})) \cdot (OVMAG + (V_{gs} - V_{ds}))] \quad (171)$$

where W_{effc} is expressed as in Eq.(6).

The overlap capacitance flag (**COOVLP** = 0) calculates bias-independent overlap capacitances. User-defined values can be specified using the input parameters **CGDO** and **CGSO**.

The overlap capacitance is

$$C_{ov} = C'_{ov} \cdot W_{effc} \cdot \mathbf{NF} \quad (172)$$

where W_{effc} is expressed as in Eq.(6). C'_{ov} is the overlap capacitance per unit length, expressed as,

$$\text{for the drain side} \quad (173)$$

$$\begin{aligned} C'_{ov} &= \mathbf{CGDO}, & \text{if } \mathbf{CGDO} \text{ is given.} \\ &= \frac{\epsilon_{ox}}{\mathbf{TFOX}} \cdot \mathbf{LOVER}, & \text{if } \mathbf{CGDO} \text{ is not given.} \end{aligned} \quad (174)$$

$$\text{for the source side} \quad (175)$$

$$\begin{aligned} C'_{ov} &= \mathbf{CGSO}, & \text{if } \mathbf{CGSO} \text{ is given.} \\ &= \frac{\epsilon_{ox}}{\mathbf{TFOX}} \cdot \mathbf{LOVER}, & \text{if } \mathbf{CGSO} \text{ is not given.} \end{aligned} \quad (176)$$

The gate-to-bulk overlap capacitance C_{gbo} is calculated only with a user-defined value $\mathbf{CGBO}$ using

$$C_{gbo} = -\mathbf{CGBO} \cdot L_{gate} \quad (177)$$

irrespective of the model Flag $\mathbf{COOVLP}$.

14.3 Gate-Fringing Capacitance

The outer fringing capacitance is modeled as [21]

$$C_{fring} = \frac{\epsilon_{ox}}{\pi/2} W_{gate} \cdot \mathbf{NF} \cdot \ln \left(1 + \frac{\mathbf{TPOLY}}{T_{FOX}} \right) \quad (178)$$

where $\mathbf{TPOLY}$ is the gate-poly thickness. This capacitance is bias independent.

Table 13: HiSIM_SOI model parameters introduced in Section 14 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

XQY	distance from junction to max field point
XQY1	V_{es} -dependence of Q_y
XQY2	L_{gate} -dependence of Q_y
CGSO	gate-to-source overlap cap
CGDO	gate-to-drain overlap cap
CGBO	gate-to-bulk overlap cap
LOVER	drain overlap length
NOVER	impurity concentration in drain overlap region
VFBOVER	flat-band voltage in overlap region
TPOLY	height of poly- S_i gate for fringing cap

15 Leakage Currents

In the following equations, V'_{bs} is set to zero for the floating body and to V_{bcs} for the body tie.

15.1 Substrate (body) Current

After some simplification we derive the well-known equation [23]

$$I_{\text{sub}} = \frac{C_1}{C_2} (\phi(y) - \phi(0)) I_{\text{ds}} \exp \left(-\frac{\lambda C_2}{\phi(y) - \phi(0)} \right) \quad (179)$$

where

$$\lambda^2 = \frac{\epsilon_{\text{Si}} X_j T_{\text{FOX}}}{\epsilon_{\text{ox}}} \quad (180)$$

and X_j is the junction depth.

The basic equation of Eq. (179) is modified so that measurements for all device sizes can be reproduced.

$$I_{\text{sub}} = \text{SUB1} \cdot \left(1 + \frac{\text{SUB1L}}{(L/10^{-6})^{\text{SUB1LP}}} \right) \cdot P_{\text{sisubsat}} \cdot I_{\text{ds}} \cdot \exp \left(-\frac{\text{SUB2} \cdot \left(1 + \frac{\text{SUB2L}}{L/10^{-6}} \right)}{P_{\text{sisubsat}}} \right) \quad (181)$$

COSUBSCALE = 0 (backward compat.)

$$P_{\text{sisubsat}} = \text{SVDS} \cdot V_{\text{ds}} + P_{\text{s0}} - \frac{1}{1 + \frac{\text{SLG}}{L/10^{-6}}} \left(1 + \frac{\text{SVGSW}}{(W/10^{-6})^{\text{SVGSWP}}} \right) \cdot \left[\text{SVGS} \cdot \left(1 + \frac{\text{SVGSL}}{(L/10^{-6})^{\text{SVGSLP}}} \right) \cdot V_{\text{g'sub}} + \frac{qN_{\text{SOI}}\epsilon_{\text{Si}}}{C_{\text{fox}}^2} \left\{ 1 - \sqrt{1 + \frac{2C_{\text{fox}}^2}{qN_{\text{SOI}}\epsilon_{\text{Si}}} \cdot \left(V_{\text{g'sub}} - \frac{1}{\beta} - x_{\text{vbs}} \cdot V_{\text{b's}} \right)} \right\} \right] \quad (182)$$

or

$$P_{\text{sisubsat}} = \text{SVDS} \cdot V_{\text{ds}} + P_{\text{s0}} - \frac{1}{1 + \frac{\text{SLG}}{L/10^{-6}}} \left(1 + \frac{\text{SVGSW}}{(W/10^{-6})^{\text{SVGSWP}}} \right) \cdot \left[\text{SVGS} \cdot \left(1 + \frac{\text{SVGSL}}{(L/10^{-6})^{\text{SVGSLP}}} \right) \cdot V_{\text{g'sub}} + \frac{qN_{\text{SOI}}\epsilon_{\text{Si}}}{C_{\text{fox}}^2} \left\{ 1 - \sqrt{1 + \frac{2C_{\text{fox}}^2}{qN_{\text{SOI}}\epsilon_{\text{Si}}} \cdot \left(V_{\text{g'sub}} - \frac{1}{\beta} - x_{\text{vbs}} \cdot \left(V_{\text{b's}} + \text{DVBSSUB} \cdot \frac{Q_{\text{h}}}{C_{\text{SOI}}} \right) \right)} \right\} \right] \quad (183)$$

for body-tie devices and **COFBE**=2.

$$V_{\text{g'sub}} = V_{\text{gs}} - \text{VFBSUB} \cdot \left(1 + \frac{\text{VFBSUBL}}{(L/10^{-6})^{\text{VFBSUBLP}}} \right) + \Delta V_{\text{th}} - \phi_{\text{Spg}} \quad (184)$$

or

$$V_{\text{g'sub}} = V_{\text{gs}} - \text{VFBSUB} \cdot \left(1 + \frac{\text{VFBSUBL}}{(L/10^{-6})^{\text{VFBSUBLP}}} \right) + \Delta V_{\text{th}} - \phi_{\text{Spg}} + \text{DVGPSUB} \quad (185)$$

for body-tie devices and **COFBE**=2.

$$x_{\text{vbs}} = \text{SVBS} \cdot \left(1 + \frac{\text{SVBSL}}{(L/10^{-6})^{\text{SVBSLP}}} \right) \quad (186)$$

COSUBSCALE = 1 (new for **HiSIM_SOI**; current for **HiSIM2** and **HiSIM_HV**.)

$$P_{\text{sisubsat}} = \mathbf{SVDS} \cdot V_{\text{ds}} + P_{\text{s0}} - \frac{1}{1 + \frac{\mathbf{SLG}}{L/10^{-6}} \left(1 + \frac{\mathbf{SLGL}}{(L/10^{-6})^{\mathbf{SLGLP}}} \right)} \cdot \left[aV_{g'\text{sub}} + \frac{qN_{\text{SOI}}\epsilon_{\text{Si}}}{C_{\text{fox}}^2} \left(1 - \sqrt{1 + \frac{2C_{\text{fox}}^2}{qN_{\text{SOI}}\epsilon_{\text{Si}}} \cdot \left(aV_{g'\text{sub}} - \frac{1}{\beta} - x_{\text{vbs}} \cdot V_{b's} \right)} \right) \right] \quad (187)$$

or

$$P_{\text{sisubsat}} = \mathbf{SVDS} \cdot V_{\text{ds}} + P_{\text{s0}} - \frac{1}{1 + \frac{\mathbf{SLG}}{L/10^{-6}} \left(1 + \frac{\mathbf{SLGL}}{(L/10^{-6})^{\mathbf{SLGLP}}} \right)} \cdot \left[aV_{g'\text{sub}} + \frac{qN_{\text{SOI}}\epsilon_{\text{Si}}}{C_{\text{fox}}^2} \left(1 - \sqrt{1 + \frac{2C_{\text{fox}}^2}{qN_{\text{SOI}}\epsilon_{\text{Si}}} \cdot \left\{ aV_{g'\text{sub}} - \frac{1}{\beta} - x_{\text{vbs}} \cdot \left(V_{b's} + \mathbf{DVBSSUB} \cdot \frac{Q_{\text{h}}}{C_{\text{SOI}}} \right) \right\}} \right) \right] \quad (188)$$

for body-tie devices and **COFBE**=2.

$$a = \mathbf{SVGS} \cdot \left(1 + \frac{\mathbf{SVGSL}}{(L/10^{-6})^{\mathbf{SVGSLP}}} \right) \cdot \frac{1}{1 + \frac{\mathbf{SVGSW}}{(W/10^{-6})^{\mathbf{SVGSWP}}}} \quad (189)$$

$$V_{g'\text{sub}} = V_{\text{gs}} - \mathbf{VFBSUB} \cdot \left(1 + \frac{\mathbf{VFBSUBL}}{(L/10^{-6})^{\mathbf{VFBSUBLP}}} \right) + \Delta V_{\text{th}} - \phi_{\text{SpG}} \quad (190)$$

or

$$V_{g'\text{sub}} = V_{\text{gs}} - \mathbf{VFBSUB} \cdot \left(1 + \frac{\mathbf{VFBSUBL}}{(L/10^{-6})^{\mathbf{VFBSUBLP}}} \right) + \Delta V_{\text{th}} - \phi_{\text{SpG}} + \mathbf{DVGPSUB} \quad (191)$$

for body-tie devices and **COFBE**=2.

$$x_{\text{vbs}} = \mathbf{SVBS} \cdot \left(1 + \frac{\mathbf{SVBSL}}{(L/10^{-6})^{\mathbf{SVBSLP}}} \right) \quad (192)$$

Impact-Ionization Induced Bulk Potential Change

The impact ionization induces electron and hole pairs, which is the origin of the substrate current. However, not only the leakage current but also the charge distribution in the bulk is changed. This induced charge redistribution affects as the bulk potential change. This is modeled in a simple way as

$$\Delta I_{\text{ds}} = \frac{2}{3} \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\left\{ \beta(\phi_{\text{sL}} - V'_{\text{bs}}) - 1 \right\}^{\frac{3}{2}} \frac{3}{2} \frac{\beta \Delta V_{\text{bulk}}}{\beta(\phi_{\text{sL}} - V'_{\text{bs}}) - 1} - \left\{ \beta(\phi_{\text{s0}} - V'_{\text{bs}}) - 1 \right\}^{\frac{3}{2}} \frac{3}{2} \frac{\beta \Delta V_{\text{bulk}}}{\beta(\phi_{\text{s0}} - V'_{\text{bs}}) - 1} \right] - \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\left\{ \beta(\phi_{\text{sL}} - V'_{\text{bs}}) - 1 \right\}^{\frac{1}{2}} \frac{1}{2} \frac{\beta \Delta V_{\text{bulk}}}{\beta(\phi_{\text{sL}} - V'_{\text{bs}}) - 1} - \left\{ \beta(\phi_{\text{s0}} - V'_{\text{bs}}) - 1 \right\}^{\frac{1}{2}} \frac{1}{2} \frac{\beta \Delta V_{\text{bulk}}}{\beta(\phi_{\text{s0}} - V'_{\text{bs}}) - 1} \right] \quad (193)$$

where

$$\Delta V_{\text{bulk}} = \mathbf{IBPC1}(1 + \mathbf{IBPC2} \cdot \Delta V_{\text{th}}) \cdot I_{\text{sub}} \quad (194)$$

IBPC1 and **IBPC2** are model parameters.

15.2 Gate Current

(i) Between Gate and Channel, I_{gate}

As for the current between gate and channel, (I_{gate}) the direct-tunneling mechanism is considered [24]. Since measured I_{gate} shows nearly linear L_{gate} dependence, the tunneling is assumed to occur along the whole channel length. Thus the final description implemented in HiSIM is [25, 26]

$$I_{\text{gate}} = q \cdot \text{GLEAK1} \cdot \frac{E^2}{E_{\text{gp}}^{\frac{1}{2}}} \cdot \exp\left(-\frac{E_{\text{gp}}^{\frac{3}{2}} \times \text{GLEAK2}}{E}\right) \cdot \sqrt{\frac{Q_i}{\text{const0}}} \cdot W_{\text{eff}} \cdot \text{NF} \cdot L_{\text{eff}} \cdot \frac{\text{GLEAK6}}{\text{GLEAK6} + V_{\text{ds}}} \cdot \frac{\text{GLEAK7}}{\text{GLEAK7} + W_{\text{eff}} \cdot \text{NF} \cdot L_{\text{eff}}} \quad (195)$$

where

$$E = \left(1 + \frac{E_y}{\text{GLEAK5}}\right) \cdot \left(1 - \frac{1}{1 + V_{\text{gs}}^2}\right) \cdot \frac{V_{\text{G}}}{T_{\text{FOX}}} \quad (196)$$

$$V_{\text{G}} = V_{\text{gs}} - V_{\text{fb}} + \text{GLEAK4} \cdot (\Delta V_{\text{th}} - \phi_{\text{SpG}}) \cdot L_{\text{eff}} - \text{GLEAK3} \cdot \phi_{\text{s}}(\Delta L) \quad (197)$$

ΔV_{th} is threshold voltage shift introduced in Section 4 (Threshold Voltage Shift), $\phi_{\text{s}}(\Delta L)$ is surface potential at the channel/drain junction and was introduced in Section 9 (Channel-Length Modulation). const0 is the same constant introduced in Section 2.

GLEAK1 through **GLEAK7** are model parameters, and E_{gp} describes the temperature dependent bandgap for the gate current (see section 12). The gate-channel current I_{gate} is partitioned into two terminal currents with one model parameter in the following manner.

$$I_{\text{gate}} = I_{\text{gate,s}} + I_{\text{gate,d}} \quad (198)$$

where

$$I_{\text{gate,s}} = (1 - P_{\text{artition}}) \cdot I_{\text{gate}} \quad (199)$$

$$I_{\text{gate,d}} = P_{\text{artition}} \cdot I_{\text{gate}} \quad (200)$$

where analytical description of P_{artition} is obtained by integrating the following equation.

$$P_{\text{artition}} = \frac{1}{I_{\text{gate}}} \int_0^{L_{\text{eff}}} \frac{y}{L_{\text{eff}}} I_{\text{gate}}(y) dy \quad (201)$$

(ii) Between Gate and Bulk, I_{gb}

The I_{gb} current under the accumulation condition is modeled as

$$I_{\text{gb}} = \text{GLKB1} \cdot E_{\text{gb}}^2 \cdot \exp\left(-\frac{\text{GLKB2}}{E_{\text{gb}}}\right) W_{\text{eff}} \cdot \text{NF} \cdot L_{\text{eff}} \quad (202)$$

$$E_{\text{gb}} = -\frac{V_{\text{gs}} - \text{VFBC} + \text{GLKB3}}{T_{\text{FOX}}} \quad (203)$$

(iii) Between Gate and Source/Drain, $I_{\text{gs}}/I_{\text{gd}}$

The tunneling current between the gate and the source/drain overlap region is modeled as

$$I_{\text{gs}} = \text{sign} \cdot \text{GLKSD1} \cdot E_{\text{gs}}^2 \exp(T_{\text{FOX}}(-\text{GLKSD2} \cdot V_{\text{gs}} + \text{GLKSD3})) W_{\text{eff}} \cdot \text{NF} \quad (204)$$

$$E_{\text{gs}} = \frac{V_{\text{gs}}}{T_{\text{FOX}}} \quad (205)$$

$$I_{\text{gd}} = \text{sign} \cdot \text{GLKSD1} \cdot E_{\text{gd}}^2 \exp(T_{\text{FOX}}(\text{GLKSD2} \cdot (-V_{\text{gs}} + V_{\text{ds}}) + \text{GLKSD3})) W_{\text{eff}} \cdot \text{NF} \quad (206)$$

$$E_{\text{gd}} = \frac{V_{\text{gs}} - V_{\text{ds}}}{T_{\text{FOX}}} \quad (207)$$

$$\begin{aligned} \text{sign} &= +1 \text{ for } E \leq 0 \\ \text{sign} &= -1 \text{ for } E \geq 0 \end{aligned}$$

15.3 GIDL (gate-induced drain leakage)

The GIDL current is generated at the drain junction under the accumulation condition. The V_{ds} increase induces a very narrow potential well in the drain just under the gate, causing carrier generation. Therefore, the GIDL current is strongly dependent on V_{ds} . At further reduced V_{gs} values the direct gate tunneling starts to dominate the I_{GIDL} measurements, resulting in V_{ds} independence. The V_{ds} dependent I_{GIDL} is modeled here. The GIDL current is modeled as a carrier generation through the direct tunneling between the valence band and the conduction band across a fairly narrow region near the drain junction.

The GIDL current is expressed as:

$$I_{GIDL} = q \cdot \mathbf{GIDL1} \cdot \frac{E^2}{E_g^{\frac{1}{2}}} \cdot \exp\left(-\mathbf{GIDL2} \cdot \frac{E_g^{\frac{3}{2}}}{E}\right) \cdot W_{eff} \cdot \mathbf{NF} \cdot \frac{V_{db}^3}{V_{db}^3 + \mathbf{GIDLVB}} \quad (208)$$

where

$$E = \frac{\mathbf{GIDL3} \cdot (V_{ds} + \mathbf{GIDL4}) - V'_G}{T_{FOX}} \quad (209)$$

and

$$V'_G = V_{gs} - \Delta V_{th} \cdot \mathbf{GIDL5} \quad (210)$$

$$V_{db} = V_{ds} - V'_{bs} \quad (211)$$

Here ΔV_{th} is defined as

$$\Delta V_{th} = \Delta V_{th,SC} + \Delta V_{th,P} \quad (212)$$

where $\Delta V_{th,SC}$ and $\Delta V_{th,P}$ are defined in Section 4. It is recommended to leave $\mathbf{GIDL4}$ at its default value, zero, because I_{GIDL} does not approach to zero even under the condition of $V_{gs} = V_{ds} = 0$.

On the source side, the GISL current is calculated with the same equation as the GIDL current described above. Finally, either of I_{GIDL} or I_{GISL} is honored accordingly by the polarity of the device.

15.4 Valence band electron tunneling current

Electrons in the channel can reach to the gate electrode through oxide from the valence band as the applied gate voltage gets higher than the band gap. In HiSIM_SOI, this tunneling current is called as I_{evb} (Electron Valence Band tunneling current) and modeled as

$$I_{evb} = \mathbf{EVB1} \cdot q \left(\frac{\phi_b}{V_{FOX}} \right) \left(\frac{2\phi_b}{V_{FOX}} - 1 \right) E_{FOX}^2 \cdot \exp\left[-\frac{\mathbf{EVB2} \left(1 - \left(1 - \frac{V_{FOX}}{\phi_b} \right)^{\frac{3}{2}} \right)}{E_{FOX}}\right] \cdot W_{eff} \cdot \mathbf{NF} \cdot L_{eff} \quad (213)$$

where ϕ_b is the barrier height for hole and is fixed to 4.12V. Here E_{FOX} and V_{FOX} are described as

$$E_{FOX} = -\frac{\mathbf{FVBS} \cdot V'_{bs} - V_{FOX} + \Delta V_{th,SC} + \Delta V_{th,P} + E_g + \mathbf{EVB3}}{T_{FOX}} \quad (214)$$

$$V_{FOX} = V'_G - \phi_{s,SOI} \quad (215)$$

where E_g is the silicon bandgap.

Table 14: HiSIM_SOI model parameters introduced in Section 15 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

Substrate current	
VFBSUB	flatband voltage for I_{sub} calculation
VFBSUBL	L_{gate} -dependence of VFBSUB
VFBSUBLP	L_{gate} -dependence of VFBSUB
SUB1	substrate current coefficient of magnitude
SUB1L	L_{gate} -dependence of SUB1
SUB1LP	L_{gate} -dependence of SUB1
SUB2	substrate current coefficient of exponential term
SUB2L	L_{gate} -dependence of SUB2
SUBDLT	smoothing parameter
SVDS	substrate current dependence on V_{ds}
SLG	substrate current dependence on L_{gate}
SLGL	substrate current dependence on L_{gate}
SLGLP	substrate current dependence on L_{gate}
SVGSW	W_{gate} -dependence of SVGS
SVGSWP	W_{gate} -dependence of SVGS
SVGS	substrate current dependence on V_{gs}
SVGSL	L_{gate} -dependence of SVGS
SVGSLP	L_{gate} -dependence of SVGS
SVBS	substrate current dependence on V_{es}
SVBSL	L_{gate} -dependence of SVBS
SVBSLP	L_{gate} -dependence of SVBS
DVGPSUB	threshold voltage shift due to (pseudo) floating body charge Qh in body-tie devices
DVBSSUB	bias shifter due to (pseudo) floating body charge Qh in body-tie devices
IBPC1	impact-ionization-induced bulk potential change
IBPC2	impact-ionization-induced bulk potential change

Gate leakage current	
GLEAK1	gate-to-channel current coefficient
GLEAK2	gate-to-channel current coefficient
GLEAK3	gate-to-channel current coefficient
GLEAK4	gate-to-channel current coefficient
GLEAK5	gate-to-channel short channel correction
GLEAK6	gate-to-channel V_{ds} -dependence correction
GLEAK7	gate-to-channel L and W dependence correction
GLKB1	gate-to-body current coefficient
GLKB2	gate-to-body current coefficient
GLKB3	gate-to-body current coefficient
GLKSD1	gate-to-S/D current coefficient
GLKSD2	gate-to-S/D current coefficient
GLKSD3	gate-to-S/D current coefficient
Gate-induced drain/source leakage	
GIDL1	magnitude of GIDL
GIDL2	field-dependence of GIDL
GIDL3	V_{ds} -dependence of GIDL
*GIDL4	threshold for V_{ds} dependence
*GIDL5	high-field correction
GIDLVB	body bias dependence coefficient
EVB1	electron tunneling from valence band
EVB2	electron tunneling from valence band
EVB3	electron tunneling from valence band
FVBS	V_{es} dependence of Fowler-Nordheim current

16 Floating-Body Effect Model

The charge storage in the SOI layer Q_h is one of the most important phenomena which causes the floating-body effect (FBE).

Model selector **COFBE** as well as **COISUB** must be set to 1 to activate FBE model. In HiSIM_SOI, Q_h is modeled as follows:

$$Q_h = \left\{ \left[\exp[-\beta(\phi_{s0,SOI} - \Delta V_{sb})] + \beta(\phi_{s0,SOI} - \Delta V_{sb}) - 1 \right]^{1/2} - \left[\exp[-\beta\phi_{s0,SOI}] + \beta\phi_{s0,SOI} - 1 \right]^{1/2} \right\} \quad (216)$$

$$\Delta V_{sb} = \mathbf{QHE1} \cdot \frac{1}{\beta} \log \left[1 + \frac{(I_{sub} + I_{evb})L_p L_n}{q \cdot \mathbf{TSOI} \cdot W_{eff} e^{-\beta \cdot \mathbf{QHE2}} (D_n N_d L_p + D_p N_d L_n)} \right] \quad (217)$$

where

$$L_n = \sqrt{D_n \cdot 10^{-7}} \quad (218)$$

$$L_p = \sqrt{D_p \cdot 10^{-7}} \quad (219)$$

$$N_d = 10^{20} \text{ cm}^{-3} \quad (220)$$

$$D_n = 36 \text{ cm}^2/\text{sec} \quad (221)$$

$$D_p = 13 \text{ cm}^2/\text{sec} \quad (222)$$

Resulting Q_h is included in the Poisson equation (see Eq. (7)) for the consistent inclusion of the FBE.

HiSIM_SOI solves Eqs. (7), (8) and (9) to find surface potentials.

The impact-ionization current and FBE calculation steps in HiSIM_SOI are summarized below with the flowchart in Fig. 12.

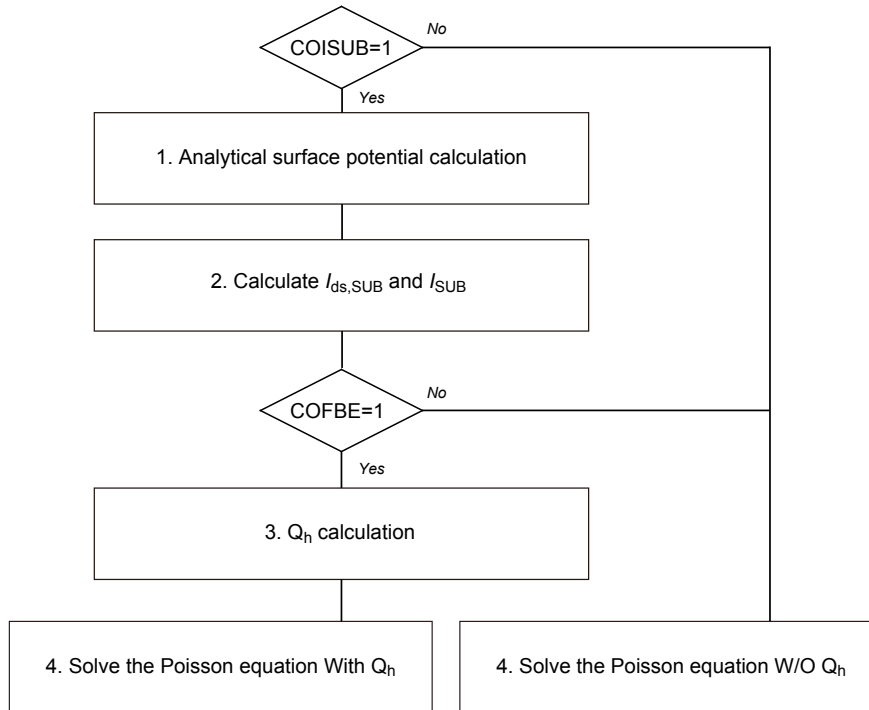


Fig. 12: Calculation flow of the impact-ionization current and floating-body effect in HiSIM_SOI

For floating-body devices (**COBCNODE**=0), the calculation of Q_h precedes the solution of the Poisson equation, and thus, it requires a preliminary estimation of surface potential, drain current and then body current. After the solution of the Poisson equation, drain current and terminal charges are readily calculated. In versions earlier than 1.5.0, body current was not reevaluated after the solution of the Poisson equation, but a newer version from 1.5.0 on, with **COISUBFB**=1,

body current is reevaluated from a fresh solution of the Poisson equation.

For body-tie devices (**COBCNODE**=1), thanks to a body-tie contact, floating body effects are usually minimal. In some advanced technologies, however, the effects are still visible on device characteristics. To account for the floating body effects on body-tie devices (**COBCNODE**=1), use **COFBE**=2 (new from 1.5.0 on), **COISUB** = 1, **COHIST**=1. The code internally stores the floating-body charge per unit area, Q_h , to an internal node reserved for this quantity. Q_h is stored and referenced in a similar manner to various charges in the NQS mode. Q_h can directly affect a solution of the Poisson equation and its subsequent calculations such as body current and floating-body effects.

The impact of Q_h on body current is, as was presented in Section 15, accounted by a corrective term such as:

$$+ \text{DVBSSUB} \cdot \frac{Q_h}{C_{\text{SOI}}} \text{ (for } P_{\text{sisubsat}} \text{ in (183) or (188))}$$

$$+ \text{DVGPSUB} \text{ (for } V_{g'_{\text{sub}}} \text{ in (185) or (191))}$$

These provide an effective shift in the back-side SOI body potential and in the effective gate voltage, both intended to make up for the lack of back-side SOI potential in the body-tie device codes (**COBCNODE**=1), while the floating-body device codes (**COBCNODE**=0) accounts for the SOI body potential through the back side body potential $\phi_{b,\text{SOI}}$ consistently and thus Q_h , without the aid of those terms. To safeguard, Q_h never exceeds **QHSMAX** in the code. Again, this safeguard mechanism also works for **COFBE** = 2 together with **COISUB** = 1 and **COBCNODE**=1 only.

The kink effect in the current can be observed with **COFBE**=1 and **COISUB**=1 as shown in Fig. 13.

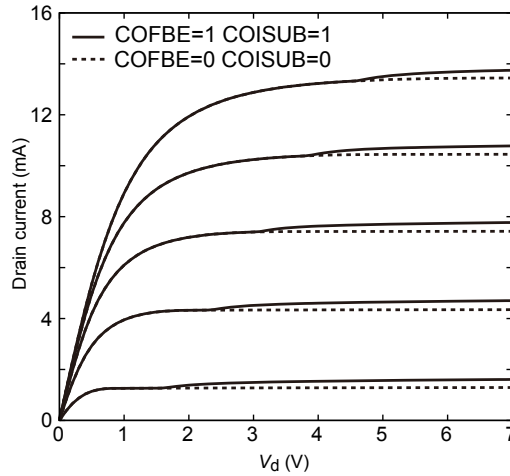


Fig. 13: Kink effect in the drain current. In this example, $L=0.6\mu\text{m}$, $W=10\mu\text{m}$, $\text{SVGS}=2.4$ and $\text{QHE2}=0.5$.

Table 17: HiSIM_SOI model parameters introduced in Section 16 of this manual. * indicates parameters that can be used for this portion of the model.

QHE1	floating-body effect parameter
QHE2	floating-body effect parameter
QHSMAX	upper-limits floating-body charge (per unit area) (body-tie devices(COBCNODE =1) and COFBE = 2)

17 History Effect Model

Time constant for the history effect is modelled as

$$\tau_h = R_{\text{sb}} \cdot C_{\text{FOX}} \quad (223)$$

$$R_{\text{sb}} = \frac{\text{HIST1}}{I_{\text{sub}} + \text{HIST2}} \quad (224)$$

where **HIST1** and **HIST2** are model parameters.

History effect is described with the same formula as the NQS effect as

$$Q_h(t) = Q_h(t - \Delta t) + \frac{\Delta t}{\tau_h + \Delta t} \cdot (Q_{h0} - Q_h(t - \Delta t)) \quad (225)$$

where Q_{h0} is the quasi-static hole storage in the SOI layer without charging delay shown in Eq. (216) in section 16.

Table 18: HiSIM_SOI model parameters introduced in Section 17 of this manual. * indicates parameters that can be used for this portion of the model.

HIST1	history-effect parameter
HIST2	history-effect parameter

18 Self-Heating Effect Model

The self-heating effect is modeled as a C-R thermal network (Fig. 14) with a heat resistance parameter **RTH0** and capacitance parameter **CTH0**. The flag **COSELFHEAT** must be equal to 1 and **RTH0** must not be equal to zero to activate this model. The SHE should be switched on/off only with the model flag **COSELFHEAT**.

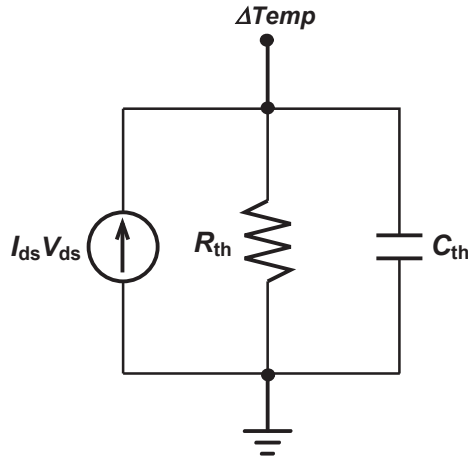


Fig. 14: Thermal Network applied for the self-heating effect. $Temp \leftarrow Temp + \Delta Temp$. $\Delta Temp$ is the incremental temperature change derived from the self-heating effect in each time step and $Temp$ is the accumulated temperature change in comparison to the initial temperature.

Table 19: HiSIM_SOI model parameters introduced in Section 18 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

RTH0	thermal resistance
CTH0	thermal capacitance

19 Source/Body and Drain/Body Diode Models

19.1 Diode Current

The model equations for the source/body and drain/body diode currents are based on the concepts of BSIM3v3 [27], but include a number of modifications.

The model for forward-biased current density is given in Eq. (227). The corresponding backward-biased current density is given in Eq. (228).

$$T_{\text{tnom}} = \frac{T}{\mathbf{TNOM}} \quad (226)$$

$$j_s = \mathbf{JS0} \cdot \exp \left\{ \frac{(E_{\text{gtnom}} \cdot \beta_{\text{tnom}} - E_g \beta + \mathbf{XTI} \cdot \log(T_{\text{tnom}}))}{\mathbf{NJ}} \right\} \quad (227)$$

$$j_{s2} = \mathbf{JS0} \cdot \exp \left\{ \frac{(E_{\text{gtnom}} \cdot \beta_{\text{tnom}} - E_g \beta + \mathbf{XTI2} \cdot \log(T_{\text{tnom}}))}{\mathbf{NJ}} \right\} \quad (228)$$

$$E_{\text{gtnom}} = \mathbf{EG0} - 90.25 \times 10^{-6} \cdot \mathbf{TNOM} - 1.0 \times 10^{-7} \cdot \mathbf{TNOM}^2 \text{ (Eq. (140) in Section 12)} \quad (229)$$

$$\beta_{\text{tnom}} = \frac{q}{k_B \cdot \mathbf{TNOM}} \text{ (Eq. (137) in Section 12)} \quad (230)$$

$$N_{\text{vtm}} = \frac{\mathbf{NJ}}{\beta} \quad (231)$$

19.1.1 Between drain and body

The forward and backward currents between drain and body are calculated as

$$I_{\text{sbd}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot \mathbf{TSOI} \cdot j_s \quad (232)$$

$$I_{\text{sbd2}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot \mathbf{TSOI} \cdot j_{s2} \quad (233)$$

The drain-body current equations are derived as follows:

$$V_{\text{bcd}} = V_{\text{bcs}} - V_{\text{ds}} \quad (234)$$

$$v_{\text{bdt}} = N_{\text{vtm}} \cdot \log \left\{ \frac{\mathbf{VDIFFJ}}{I_{\text{sbd}}} \cdot (T_{\text{tnom}})^2 + 1 \right\} \quad (235)$$

(i) $V_{\text{bcd}} < v_{\text{bdt}}$

$$I_{\text{bd}} = I_{\text{sbd}} \left\{ \exp \left(\frac{V_{\text{bcd}}}{N_{\text{vtm}}} \right) - 1 \right\} \quad (236)$$

(ii) $V_{\text{bcd}} \geq v_{\text{bdt}}$

$$I_{\text{bd}} = I_{\text{sbd}} \left\{ \exp \left(\frac{v_{\text{bdt}}}{N_{\text{vtm}}} \right) - 1 \right\} + \frac{I_{\text{sbd}}}{N_{\text{vtm}}} \exp \left(\frac{v_{\text{bdt}}}{N_{\text{vtm}}} \right) (V_{\text{bcd}} - v_{\text{bdt}}) \quad (237)$$

$$I_{\text{bd}} = I_{\text{bd}} + \mathbf{DIVX} \cdot I_{\text{sbd2}} \cdot V_{\text{bcd}} \quad (238)$$

19.1.2 Between source and body

$$I_{\text{sbs}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot \mathbf{TSOI} \cdot j_s \quad (239)$$

$$I_{\text{sbs2}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot \mathbf{TSOI} \cdot j_{s2} \quad (240)$$

$$v_{\text{bst}} = N_{\text{vtm}} \cdot \log \left\{ \frac{\mathbf{VDIFFJ}}{I_{\text{sbs}}} \cdot (T_{\text{tnom}})^2 + 1 \right\} \quad (241)$$

(i) $V_{\text{bcs}} < v_{\text{bst}}$

$$I_{\text{bs}} = I_{\text{sbs}} \cdot \left\{ \exp \left(\frac{V_{\text{bcs}}}{N_{\text{vtm}}} \right) - 1 \right\} \quad (242)$$

(ii) $V_{\text{bcs}} \geq v_{\text{bst}}$

$$I_{\text{bs}} = I_{\text{sbs}} \left\{ \exp \left(\frac{v_{\text{bst}}}{N_{\text{vtm}}} \right) - 1 \right\} + \frac{I_{\text{sbs}}}{N_{\text{vtm}}} \exp \left(\frac{v_{\text{bst}}}{N_{\text{vtm}}} \right) (V_{\text{bcs}} - v_{\text{bst}}) \quad (243)$$

$$I_{\text{bs}} = I_{\text{bs}} + \mathbf{DIVX} \cdot I_{\text{sbs2}} \cdot V_{\text{bcs}} \quad (244)$$

The hard-breakdown model of the diode will be implemented in a future version of HiSIM_SOI.

19.2 Diode Capacitance

The diode capacitances, C_{apbs} for the source/body junction and C_{apbd} for the drain/body junction, are given by the following equations. These equations have the same basis as those used in BSIM3v3 [27], but include a number of minor modifications.

The notations $\Theta = \text{S}, \theta = \text{s}$ (for source/body junction) and $\Theta = \text{D}, \theta = \text{d}$ (for drain/body junction) apply

–Suggested rewrite– (2018.2.21)

For brevity, the following notations are used in this subsection. Generic symbols should be read as source-side and drain-side specific ones respectively, when source-side diode capacitance and drain-side diode capacitance are addressed.

Generic symbol	Source-side	Drain-side	
$\mathbf{A\Theta}$	$\mathbf{AS}$	$\mathbf{AD}$	junction area (instance parameter)
$\mathbf{P\Theta}$	$\mathbf{PS}$	$\mathbf{PD}$	junction peripheral length (instance parameter)
$V_{\text{bc}\theta}$	V_{bcs}	V_{bcd}	voltage across diode (between body and source or drain)
$C_{\text{zb}\theta}$	C_{zbs}	C_{zbd}	area component of diode capacitance
$C_{\text{zb}\theta\text{sw}}$	C_{zbsw}	C_{zbdsw}	sidewall periphery component of diode capacitance
$C_{\text{zb}\theta\text{swg}}$	C_{zbswg}	C_{zbdswg}	gate-side periphery component of diode capacitance
$Q_{\text{b}\theta}$	Q_{bs}	Q_{bd}	diode capacitance charge
$C_{\text{apb}\theta}$	C_{apbs}	C_{apbd}	diode capacitance

First of all, the area component of diode capacitance is expressed as

$$c_{\text{zb}\theta} = \mathbf{CJ} \cdot \mathbf{A\Theta} \quad (245)$$

where $\mathbf{A\Theta} = 0$, if $(\text{TSOI} - \mathbf{XJ}) \leq 0$.

(I) $\mathbf{P\Theta} > W_{\text{eff}}$

$$c_{\text{zb}\theta\text{sw}} = \mathbf{CJSW}(\mathbf{P\Theta} - W_{\text{effc}} \cdot \mathbf{NF}) \quad (246)$$

$$c_{\text{zb}\theta\text{swg}} = \mathbf{CJSWG} \cdot W_{\text{effc}} \cdot \mathbf{NF} \quad (247)$$

(i) $V_{\text{bc}\theta} = 0$

$$Q_{\text{b}\theta} = 0 \quad (248)$$

$$C_{\text{apb}\theta} = c_{\text{zb}\theta} + c_{\text{zb}\theta\text{sw}} + c_{\text{zb}\theta\text{swg}} \quad (249)$$

(ii) $V_{\text{bc}\theta} < 0$

a-1) $c_{\text{zb}\theta} > 0$

$$\arg = 1 - \frac{V_{\text{bc}\theta}}{\mathbf{PB}} \quad (250)$$

α) $\mathbf{MJ} = 0.5$

$$\text{sarg} = \frac{1}{\sqrt{\arg}} \quad (251)$$

β) $\mathbf{MJ} \neq 0.5$

$$\text{sarg} = \exp(-\mathbf{MJ} \cdot \log(\arg)) \quad (252)$$

$$Q_{\text{b}\theta} = \frac{\mathbf{PB} \cdot c_{\text{zb}\theta}(1 - \arg \cdot \text{sarg})}{1 - \mathbf{MJ}} \quad (253)$$

$$C_{\text{apb}\theta} = c_{\text{zb}\theta} \cdot \text{sarg} \quad (254)$$

In SOI-MOSFET, S/D(bottom part) - body diode charges ($Q_{\text{b}\theta}$) are limited by the BOX. In HiSIM_SOI, the maximum depletion charges are expressed as

$$Q_{\text{b}\theta\text{max}} = q \cdot N_{\text{subs}} (\text{TSOI} - \mathbf{XJ}) \cdot \mathbf{A\Theta} \quad (255)$$

$$\text{a-2) } c_{zb\theta} \leq 0$$

$$Q_{b\theta} = 0 \quad (256)$$

$$C_{apb\theta} = 0 \quad (257)$$

$$\text{b) } c_{zb\theta_{sw}} > 0$$

$$\arg = 1 - \frac{V_{bc\theta}}{\mathbf{PBSW}} \quad (258)$$

$$\alpha) \mathbf{MJSW} = 0.5$$

$$\text{sarg} = \frac{1}{\sqrt{\arg}} \quad (259)$$

$$\beta) \mathbf{MJSW} \neq 0.5$$

$$\text{sarg} = \exp(-\mathbf{MJSW} \cdot \log(\arg)) \quad (260)$$

$$\begin{aligned} Q_{b\theta} &= Q_{b\theta} + \frac{\mathbf{PBSW} \cdot c_{zb\theta_{sw}}(1 - \arg \cdot \text{sarg})}{1.0 - \mathbf{MJSW}} \\ C_{apb\theta} &= C_{apb\theta} + c_{zb\theta_{sw}} \cdot \text{sarg} \end{aligned} \quad (262)$$

$$\text{c) } c_{zb\theta_{swg}} > 0$$

$$\arg = 1 - \frac{V_{bc\theta}}{\mathbf{PBSWG}} \quad (263)$$

$$\alpha) \mathbf{MJSWG} = 0.5$$

$$\text{sarg} = \frac{1}{\sqrt{\arg}} \quad (264)$$

$$\beta) \mathbf{MJSWG} \neq 0.5$$

$$\text{sarg} = \exp(-\mathbf{MJSWG} \cdot \log(\arg)) \quad (265)$$

$$\begin{aligned} Q_{b\theta} &= Q_{b\theta} + \frac{\mathbf{PBSWG} \cdot c_{zb\theta_{swg}}(1 - \arg \cdot \text{sarg})}{1 - \mathbf{MJSWG}} \\ C_{apb\theta} &= C_{apb\theta} + c_{zb\theta_{swg}} \cdot \text{sarg} \end{aligned} \quad (267)$$

$$\text{(iii) } V_{bc\theta} > 0$$

$$\begin{aligned} Q_{b\theta} &= V_{bc\theta}(c_{zb\theta} + c_{zb\theta_{sw}} + c_{zb\theta_{swg}}) \\ &+ V_{bc\theta}^2 \left(\frac{1}{2} \frac{c_{zb\theta} \cdot \mathbf{MJ}}{\mathbf{PB}} + \frac{1}{2} \frac{c_{zb\theta_{sw}} \cdot \mathbf{MJSW}}{\mathbf{PBSW}} + \frac{1}{2} \frac{c_{zb\theta_{swg}} \cdot \mathbf{MJSWG}}{\mathbf{PBSWG}} \right) \end{aligned} \quad (268)$$

$$\begin{aligned} C_{apb\theta} &= c_{zb\theta} + c_{zb\theta_{sw}} + c_{zb\theta_{swg}} \\ &+ V_{bc\theta} \left(\frac{c_{zb\theta} \cdot \mathbf{MJ}}{\mathbf{PB}} + \frac{c_{zb\theta_{sw}} \cdot \mathbf{MJSW}}{\mathbf{PBSW}} + \frac{c_{zb\theta_{swg}} \cdot \mathbf{MJSWG}}{\mathbf{PBSWG}} \right) \end{aligned} \quad (269)$$

$$\text{(II) } \mathbf{P}\Theta \leq W_{\text{eff}}$$

$$c_{zb\theta_{swg}} = \mathbf{CJSWG} \cdot \mathbf{P}\Theta \quad (270)$$

$$\text{(i) } V_{bc\theta} = 0$$

$$Q_{b\theta} = 0 \quad (271)$$

$$C_{apb\theta} = c_{zb\theta} + c_{zb\theta_{swg}} \quad (272)$$

$$\text{(ii) } V_{bc\theta} < 0$$

$$\text{a-1) } c_{zb\theta} > 0$$

$$\arg = 1 - \frac{V_{bc\theta}}{\mathbf{PB}} \quad (273)$$

$$\alpha) \mathbf{MJ} = 0.5$$

$$\text{sarg} = \frac{1}{\sqrt{\text{arg}}} \quad (274)$$

$$\beta) \mathbf{MJ} \neq 0.5$$

$$\text{sarg} = \exp(-\mathbf{MJ} \cdot \log(\text{arg})) \quad (275)$$

$$Q_{b\theta} = \frac{\mathbf{PB} \cdot c_{zb\theta}(1 - \text{arg} \cdot \text{sarg})}{1 - \mathbf{MJ}} \quad (276)$$

$$C_{apb\theta} = c_{zb\theta} \cdot \text{sarg} \quad (277)$$

$$\text{a-2) } c_{zb\theta} \leq 0$$

$$Q_{b\theta} = 0 \quad (278)$$

$$C_{apb\theta} = 0 \quad (279)$$

$$\text{b) } c_{zb\theta\text{swg}} > 0$$

$$\text{arg} = 1 - \frac{V_{bc\theta}}{\mathbf{PBSWG}} \quad (280)$$

$$\alpha) \mathbf{MJSWG} = 0.5$$

$$\text{sarg} = \frac{1}{\sqrt{\text{arg}}} \quad (281)$$

$$\beta) \mathbf{MJSWG} \neq 0.5$$

$$\text{sarg} = \exp(-\mathbf{MJSWG} \cdot \log(\text{arg})) \quad (282)$$

$$Q_{b\theta} = Q_{b\theta} + \mathbf{PBSWG} \cdot c_{zb\theta\text{swg}} \cdot \frac{1 - \text{arg} \cdot \text{sarg}}{1 - \mathbf{MJSWG}} \quad (283)$$

$$C_{apb\theta} = C_{apb\theta} + c_{zb\theta\text{swg}} \cdot \text{sarg} \quad (284)$$

$$\text{(iii) } V_{bc\theta} > 0$$

$$Q_{b\theta} = V_{bc\theta} \cdot (c_{zb\theta} + c_{zb\theta\text{swg}}) + V_{bc\theta}^2 \left(\frac{1}{2} \frac{c_{zb\theta} \cdot \mathbf{MJ}}{\mathbf{PB}} + \frac{1}{2} \frac{c_{zb\theta\text{swg}} \cdot \mathbf{MJSWG}}{\mathbf{PBSWG}} \right) \quad (285)$$

$$C_{apb\theta} = c_{zb\theta} + c_{zb\theta\text{swg}} + V_{bc\theta} \left(\frac{c_{zb\theta} \cdot \mathbf{MJ}}{\mathbf{PB}} + \frac{c_{zb\theta\text{swg}} \cdot \mathbf{MJSWG}}{\mathbf{PBSWG}} \right) \quad (286)$$

For the fully depleted case, the bottom capacitance becomes zero. Even for the PD condtion, the diode depletion stretches to BOX under the reverse mode, where the bottom capacitance becomes zero. These condtions are included.

Table 21: HiSIM_SOI model parameters introduced in Section 21 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

XTI	temp coefficient for forward current densities
XTI2	temp coefficient for forward current densities
NJ	emission coefficient
VDIFFJ	diode threshold voltage at junction
DIVX	reverse current coefficient
CJ	bottom junc cap/unit area at zero bias
CJSW	sidewall junc cap grading coefficient
CJSWG	sidewall junc cap per unit length at zero bias
PB	bottom junc built-in potential
PBSW	sidewall junc built-in potential
PBSWG	gate sidewall junc built-in potential
MJ	bottom junc cap grading coefficient
MJSW	sidewall junc cap grading coefficient
MJSWG	gate sidewall junc cap grading coefficient

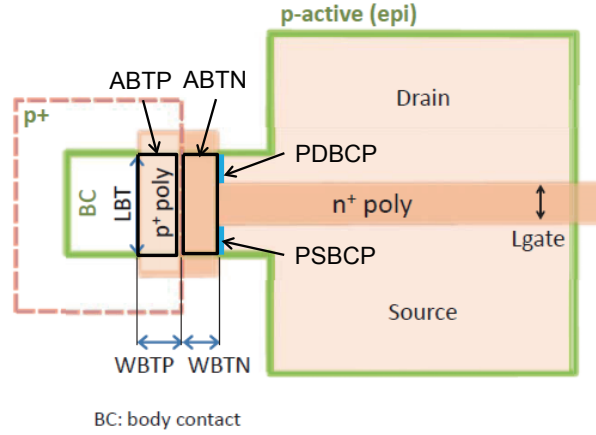


Fig. 15: Instance parameters for body contact MOS capacitance for the nMOS-type.

20 Body-Tie Models

HiSIM_SOI considers both the floating-body and the body-tie structures. Modeling of effects induced by the additional area of the body-tie-contact region is described here.

20.1 Body-Contact Capacitances

Additional capacitance induced by the body contact is modeled here. Instance parameters for body-tied MOS capacitance are shown in Fig. 15 for the nMOS type.

The instance parameters **LBT**, **WBTN**, and **WBTP** are used for area calculations of respective capacitor. The body-contact capacitances are activated by defining **WBTN** and **WBTP** or by defining **ABTN** and **ABTP** (see Fig. 15). If **ABTN** and **ABTP** are not defined, they are calculated as

$$ABTN = LBT \cdot WBTN \quad (287)$$

$$ABTP = LBT \cdot WBTP \quad (288)$$

20.1.1 Overlap capacitance of body-tie area

The capacitance between the body-tied gate (hammer head) and the body-tied well is treated as MOS capacitance

$$C_{gb, bt} = C_{btN} \cdot ABTN + C_{btP} \cdot ABTP \quad (289)$$

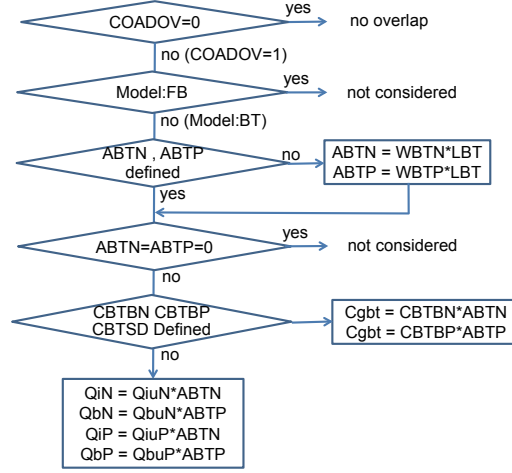


Fig. 16: Criteria for the body-tied capacitance calculation

where C_{btN} and C_{btP} are the gate capacitances over the body-tie well per unit area. C_{btN} and C_{btP} are

$$C_{btN} = \text{CBTBN}, \quad \text{if CBTBN is given.} \quad (290)$$

$$= \text{calculated by charges induced in MOSFET with } n^+ \text{ poly,} \quad \text{if CBTBN is not given.} \quad (291)$$

$$C_{btP} = \text{CBTBP}, \quad \text{if CBTBP is given.} \quad (292)$$

$$= \text{calculated by charges induced in MOSFET with } p^+ \text{ poly,} \quad \text{if CBTBP is not given.} \quad (293)$$

20.1.2 Overlap capacitance of body-tie periphery

Instance parameters **PSBCP** and **PDBCP** (see Fig. 15) are introduced to take into account the overlap capacitance of the periphery of the body-tie contact. These lengths are added to the W_{effc} as

$$W_{\text{effc,source}} = W_{\text{effc}} + \text{PSBCP} \quad (294)$$

$$W_{\text{effc,drain}} = W_{\text{effc}} + \text{PDBCP} \quad (295)$$

where

$$W_{\text{effc}} = W_{\text{gate}} - \text{NBT} \cdot \text{XWDBT} - (2 - \text{NBT}) \cdot \text{XWDC} \quad (296)$$

If **XWDC** is not defined, the channel width for the capacitance is equal to that for the current.

The effective channel width $W_{\text{effc,source}}$ and $W_{\text{effc,drain}}$ are used for calculating the overlap capacitance described in Section 14.2.

20.1.3 Diode current of body-tie periphery

The diode current also flows at the body-tie periphery. Therefore the lengths determined by the instance parameters **PSBCP** and **PDBCP** are added the width as

$$W_{\text{eff,source}} = W_{\text{eff}} + \text{PSBCP} \quad (297)$$

$$W_{\text{eff,drain}} = W_{\text{eff}} + \text{PDBCP} \quad (298)$$

where

$$W_{\text{eff}} = W_{\text{gate}} - \text{NBT} \cdot \text{XWDBT} - (2 - \text{NBT}) \cdot \text{XWD} \quad (299)$$

The effective channel $W_{\text{eff,source}}$ and $W_{\text{eff,drain}}$ is used in the diode current model described in Section 19.

20.1.4 Gate-Fringing Capacitance

For body-tied devices, an additional gate-fringing capacitance is considered between gate and body:

$$C_{\text{fring,bt}} = \frac{\epsilon_{\text{ox}}}{\pi/2} \text{LBT} \cdot \text{NF} \cdot \ln \left(1 + \frac{\text{TPOLY}}{T_{\text{FOX}}} \right) \quad (300)$$

Further fringing capacitacnes are added into the intrinsic part of C_{fringing} as

$$C_{\text{fringing}} = C_{\text{fringing}} \quad (301)$$

$$+ \frac{\epsilon_{\text{ox}}}{\pi/2} \text{PSBCP} \cdot \text{NF} \cdot \ln \left(1 + \frac{\text{TPOLY}}{T_{\text{FOX}}} \right) \quad (302)$$

$$+ \frac{\epsilon_{\text{ox}}}{\pi/2} \text{PDBCP} \cdot \text{NF} \cdot \ln \left(1 + \frac{\text{TPOLY}}{T_{\text{FOX}}} \right) \quad (303)$$

Table 22: HiSIM_SOI model parameters introduced in Section 21 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NBT	Number of body contact
LBT	Length of gate over body-contact well
WBTN	Distance of gate protrusion over body-contact well
WBTP	Distance of gate protrusion over body-contact well
ABTN	Area of N^+ poly area of body contact
ABTP	Area of P^+ poly area of body contact
CBTBN	N^+ poly capacitance
CBTBP	P^+ poly capacitanc
PSBCP	Parasitic perimeter length for body contact at source
PDBCP	Parasitic perimeter length for body contact at drain
XWDBT	BT overlap width

21 Noise Models

21.1 $1/f$ Noise Model

The $1/f$ noise is caused by both the carrier fluctuation and the mobility fluctuation [28]

$$S_{I_{ds}} = \frac{I_{ds}^2 \mathbf{NFTRP}}{\beta f \mathbf{FALPH} (L_{eff} - \Delta L) W_{eff} \cdot \mathbf{NF}} \left[\frac{1}{(N_0 + N^*)(N_L + N^*)} + \frac{2\mu E_y \mathbf{NFALP}}{N_L - N_0} \ln \left(\frac{N_L + N^*}{N_0 + N^*} \right) + (\mu E_y \mathbf{NFALP})^2 \right] \quad (304)$$

where the parameters $\mathbf{NFALP}$ and $\mathbf{NFTRP}$ represent the contribution of the mobility fluctuation and the ratio of trap density to attenuation coefficient, respectively. N_0 and N_L are carrier densities at source side and drain side or pinch-off point, respectively, as calculated in HiSIM. N^* is written as

$$N^* = \frac{C_{FOX} + C_{dep} + \mathbf{CIT}}{q\beta} \quad (305)$$

where C_{dep} is the depletion capacitance calculated with ϕ_s . $\mathbf{CIT}$ is the capacitance caused by the interface-trapped carriers and is normally fixed to be zero.

$$N_{flick} = S_{I_{ds}} \cdot f \quad (306)$$

is calculated in HiSIM.

21.2 Thermal Noise Model

The equations for S_{id} in our compact-modeling approach, become functions of the self-consistent surface potentials as well as the surface-potential derivatives at source and drain [29, 30].

$$S_{id} = 4kT \frac{W_{eff} \cdot \mathbf{NF} \cdot C_{FOX} V_{gvt} \mu_f}{(L_{eff} - \Delta L)} \frac{(1 + 3\eta + 6\eta^2)\mu_d^2 + (3 + 4\eta + 3\eta^2)\mu_d \mu_f + (6 + 3\eta + \eta^2)\mu_f}{15(1 + \eta)\mu_{av}^2} \quad (307)$$

with

$$\mu_d = \left(1.0 + \left(\frac{\mu_0 E_{yd}}{V_{max,therm}} \right)^{BB} \right)^{\frac{1}{BB}} \quad (308)$$

$$\mu_{av} = \frac{\mu_f + \mu_d}{2.0} \quad (309)$$

where μ_f , μ_d , and μ_{av} are mobilities at the source side (see Section 8), the drain side, averaged, respectively. E_{yd} is derived from calculated ϕ_s values. $V_{max,therm}$ is maximum electron-saturation velocity (1.0×10^7 cm/s). V_{gvt} denotes an effective gate voltage that is equal to the charge density at the source side Q_{i0} divided by the oxide capacitance C_{FOX} . const0 is the same constant introduced in Section 2.

The quantity η in (307) is calculated as

$$\eta = 1 - \frac{(\phi_{sL,SOI} - \phi_{s0,SOI}) + \chi(\phi_{sL,SOI} - \phi_{s0,SOI})}{V_{gvt}} \quad (310)$$

$$\chi = 2 \frac{\text{const0}}{C_{FOX}} \left[\left[\frac{2}{3} \frac{1}{\beta} \frac{\{\beta(\phi_{sL,SOI} - V'_{bs}) - 1\}^{\frac{3}{2}} - \{\beta(\phi_{s0,SOI} - V'_{bs}) - 1\}^{\frac{3}{2}}}{\phi_{sL,SOI} - \phi_{s0,SOI}} \right] - \sqrt{\beta(\phi_{s0,SOI} - V'_{bs}) - 1} \right] \quad (311)$$

V'_{bs} equals to 0 for floating-body devices and equals to V_{bcs} for body-tie devices.

Thus no additional model parameters are required for the thermal noise model.

$$N_{thrm} = S_{id}/4kT \quad (312)$$

is calculated in HiSIM.

21.3 Induced Gate Noise Model

No additional model parameters are required for the induced gate noise model.

$$N_{\text{igate}} = S_{\text{igate}}/f^2 \quad (313)$$

is calculated in HiSIM. Explicit model equation is presented in [31].

21.4 Coupling Noise Model

No additional model parameters are required for the coupling noise model.

$$N_{\text{cross}} = \frac{S_{\text{igid}}}{\sqrt{S_{\text{igate}} \cdot S_{\text{id}}}} \quad (314)$$

is calculated in HiSIM. Explicit model equation is in [31].

Table 23: HiSIM_SOI model parameters introduced in Section 21 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NFTRP	ratio of trap density to attenuation coefficient
NFALP	contribution of mobility fluctuation
*CIT	cap caused by the interface trapped carriers

22 Non-Quasi-Static (NQS) Model

22.1 Carrier Formation

Carriers in the channel take time to build-up as opposed to the Quasi-Static (QS) approximation. To consider this phenomenon in HiSIM, the carrier formation is modeled as [32, 33]

$$q(t_i) = \frac{q(t_{i-1}) + \frac{\Delta t}{\tau} Q(t_i)}{1 + \frac{\Delta t}{\tau}} \quad (315)$$

where $q(t_i)$ and $Q(t_i)$ represent the non-quasi-static and the quasi-static carrier density at time t_i , respectively, and $\Delta t = t_i - t_{i-1}$ is valid. Eq. (315) implies that the formation of carriers under the NQS approximation is always delayed in comparison to the QS approximation, which is the basic origin of the NQS effect. The delay is determined by the carrier transit delay τ and the time interval in the circuit simulation Δt .

22.2 Delay Mechanisms

Two charges, one is the inversion charge and the other is the bulk charge, are considered. The transit delay of Q_i is τ and that of Q_b is τ_B .

Up to weak inversion, carriers diffuse to the channel and the transit delay is approximated by

$$\tau_{\text{diff}} = \mathbf{DLY1} \quad (316)$$

At strong inversion, there is already conduction due to field-driven carriers. The transit delay due to the flow of such conductive carriers is

$$\tau_{\text{cond}} = \mathbf{DLY2} \cdot \frac{Q_i}{I_{\text{ds}}} \quad (317)$$

where **DLY2** is a constant coefficient. These two delay mechanisms (diffusion and conduction) are combined using the Matthiessen rule

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{diff}}} + \frac{1}{\tau_{\text{cond}}} \quad (318)$$

Applying the same approach for the formation of body carriers, leads to the approximation of the body carrier delay as an RC delay in the form

$$\tau_B = \mathbf{DLY3} \cdot C_{\text{FOX}} \quad (319)$$

where **DLY3** is a constant coefficient and C_{FOX} is the oxide capacitance.

22.3 Time-Domain Analysis

The total drain/source/body terminal currents are derived from the superposition of the transport current and the charging current. The transport current is a function of the instantaneous terminal voltages and is approximated by the steady-state solution. The source/drain/body charging currents are the time derivatives of the associated non-quasi-static charges, q_s , q_D , and q_B , respectively.

22.4 AC Analysis

The non-quasi-static charge formulation in the time-domain is transformed into the frequency-domain using the Fourier transformation. The resulting NQS charge $\hat{q}$ is derived as [34]

$$\hat{q}_a(\omega) = \left\{ \frac{1}{1 + (\tau\omega)^2} - i \frac{\tau\omega}{1 + (\tau\omega)^2} \right\} \hat{Q}_a(\omega) \quad (320)$$

where $\hat{Q}$ is the QS charge in the frequency-domain and τ is the carrier transit delay. The NQS capacitances are calculated as

$$\begin{aligned} C_{\text{ab}} \equiv \frac{\partial q_a}{\partial V_b} = & -\frac{2(\tau\omega)^2}{(1 + (\tau\omega)^2)^2} \frac{1}{\tau} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,\text{QS}} + \frac{1}{1 + (\tau\omega)^2} \cdot C_{\text{ab},\text{QS}} \\ & -i \left[\frac{\tau\omega(1 - (\tau\omega)^2)}{(1 + (\tau\omega)^2)^2} \frac{1}{\tau} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,\text{QS}} + \frac{\tau\omega}{1 + (\tau\omega)^2} \cdot C_{\text{ab},\text{QS}} \right] \end{aligned} \quad (321)$$

where $C_{ab,QS}$ is the quasi-static capacitance. a and b denote the MOSFET terminals; gate g , drain d , source s and bulk b . For y -parameter calculation, conventional circuit simulators implement the quasi-static description as given for example in [16]. In order to obtain the NQS y -parameters, the capacitances in the conventional equations are replaced by the NQS capacitances derived in Eq. (321).

In general, y_{ab} is obtained as

$$y_{ab} = \frac{i\omega}{1 + (\tau\omega)^2} [C_{ab,QS} + (\tau\omega)^2 A_{ab}(\omega) - i[\tau\omega B_{ab}(\omega) + \tau\omega C_{ab,QS}]] \quad (322)$$

where A_{ab} and B_{ab} are defined as

$$A_{ab} = \frac{1}{\tau} \frac{-2}{1 + (\tau\omega)^2} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} \quad (323)$$

$$B_{ab} = \frac{1}{\tau} \frac{1 - (\tau\omega)^2}{1 + (\tau\omega)^2} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} \quad (324)$$

y_{gg} , for example, is explicitly is written as

$$y_{gg} = \frac{i\omega}{1 + (\tau\omega)^2} [C_{gg,QS} + (\tau\omega)^2 A_{gg}(\omega) - i[\tau\omega B_{gg}(\omega) + \tau\omega C_{gg,QS}]] \quad (325)$$

Due to the described approach, the non-quasi static AC-analysis model implemented in HiSIM requires no additional model parameters.

Table 24: HiSIM_SOI model parameters introduced in Section 22 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

DLY1	coefficient for delay due to diffusion of carriers
DLY2	coefficient for delay due to conduction of carriers
DLY3	coefficient for RC delay of bulk carriers

23 Multiplication Factor

For the case of a multiple device construction the multiple factor M is introduced as an instance parameter. All device features such as currents and capacitances are multiplied by M . Noises are also multiplied by M .

Table 25: HiSIM_SOI model parameters introduced in Section 23 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

M	multiplication factor
-----	-----------------------

24 Conservation of Symmetry at $V_{ds} = 0$

HiSIM_SOI preserves the symmetry at $V_{ds} = 0$ automatically due to the drift-diffusion approximation. However, modeling of the short-channel effects induces a small asymmetry. To eliminate the asymmetry caused by the artifacts of the modeling, the V_{th} modeling has to include a damping of the short-channel effects as V_{ds} approaches zero. This corresponds to a vanishing of short-channel effects as V_{ds} gets near to zero, which is actually also observed in 2D simulations. In HiSIM_SOI the damping is done by a mathematical function with two parameters: **VZADD0** and **PZADD0**. The values of these parameters are fixed, and it is recommended not to change them.

Table 26: HiSIM_SOI model parameters introduced in Section 24 of this manual.

VZADD0	symmetry conservation coefficient
PZADD0	symmetry conservation coefficient

25 Flags for Model Selection

25.1 Disabling models by setting model parameters

To exclude specific modeled effects, the following parameter settings should be chosen.

Short-channel effect	SC1 = SC2 = SC3 = 0
Reverse-short-channel effect	LP = 0
Quantum-mechanical effect	QME1 = QME3 = 0
Channel-length modulation	CLM1 = CLM2 = CLM3 = 0
Narrow-channel effect	WFC = MUEPHW = 0
Small-size effect	WL2 = 0

25.2 Model selection flags

Following flags are prepared to select required model options.

1. Overlap capacitance model is selected as:
COOVLP = 0: constant overlap capacitance (default)
COOVLP = 1: bias dependent overlap capacitance based on surface potential calculation
2. Overlap capacitance model selector (continued):
COQOVSM = 0: analytical evaluation only (iterative solution for surface potential not to be invoked)
COQOVSM = 1: iteratively calculates surface potential values for evaluating overlap capacitances
3. Gate/body contact well MOS-capacitance model selector (body-contact devices or body-tie devices only):
COQBDSM = 0: analytical evaluation only (iterative solution for surface potential not to be invoked)
COQBDSM = 1: iteratively calculates surface potential values for evaluating MOS-capacitance.

Note: Setting nonzero values for either or both instance parameters (**WBTN**, **WBTP**) automatically activates the MOS-capacitance evaluation. **COQBDSM** controls the way of evaluation.

4. Substrate current I_{sub} is calculated:
COISUB = 0: no (default)
COISUB = 1: yes
5. Gate current I_{gate} is calculated:
COIIGS = 0: no (default)
COIIGS = 1: yes
6. GIDL current I_{GIDL} is calculated:
COGIDL = 0: no (default)
COGIDL = 1: yes
7. STI leakage current $I_{\text{ds,STI}}$ is calculated:
COISTI = 0: no (default)
COISTI = 1: yes
8. Lateral field induced and overlap charges/capacitances are added to intrinsic ones:
COADOV = 0: no
COADOV = 1: yes (default)
9. Non-quasi-static mode is invoked:
CONQS = 0: no (default)
CONQS = 1: yes
10. Drain resistance is included (This flag can also be given as a instance parameter):
CORD = 0: no (default)
CORD = 1: yes

11. Source resistance is included (This flag can also be given as a instance parameter):
CORS = 0: no (default)
CORS = 1: yes
12. Gate-contact resistance is included (This flag can also be given as a instance parameter):
CORG = 0: no (default)
CORG = 1: yes
13. Body resistance network is invoked:
CORBNET = 0: no (default)
CORBNET = 1: yes
14. $1/f$ noise is calculated:
COFLICK = 0: no (default)
COFLICK = 1: yes
15. Thermal noise is calculated:
COTHRML = 0: no (default)
COTHRML = 1: yes
16. Induced gate and cross correlation noise are calculated:
COIGN = 0 || **COTHRML** = 0: no (default)
COIGN = 1 & **COTHRML** = 1: yes
17. Previous ϕ_s is used for the iteration:
COPPRV = 0: no (default)
COPPRV = 1: yes
18. Self-Heating
COSELFHEAT = 0: no (default)
COSELFHEAT = 1: yes
19. Floating-Body Effect
COFBE = 0: no (default)
COFBE = 1: yes (**COISUB**=1 is also required)
COFBE = 2: yes (specific for body-tie devices; use this option together with **COBCNODE**=1, **COISUB**=1, and **COHIST**=1).
20. History Effect
COHIST = 0: no (default)
COHIST = 1: yes
21. Valence-band-electron-tunneling-current model
COIEVB = 0: no (default)
COIEVB = 1: yes
22. Symmetry treatment
COVBSBIZ = 0: no (default)
COVBSBIZ = 1: yes
23. Using L_{eff} instead of L_{gate} in following sections
Physically L_{eff} is reasonable: Section 4, 10.3, 14.2, 15.1
Just for fitting capability: Section 2, 5, 6, 8, 9, 10.1, 10.4, 14.1, 15.1
COLGLEFF = 0: no (default)
COLGLEFF = 1: yes
24. Backward compatibility of HiSIM_SOI 1.1.1
COOLDVER = 0: no (default)
COOLDVER = 1: yes
25. 5th node model selector
COBCNODE = 0: floating body (default)
COBCNODE = 1: body tie

- 26. punchthrough model selector
 - COPT** = 0: no (default)
 - COPT** = 1: punchthrough effects

26 Model Parameters

26.1 List of model parameters, default values, and ranges

The maximum and minimum limits of the model parameter are recommended values. * indicates minor model parameters.

Parameter	Unit	Min	Default	Max	Description	See
Control parameters						
VERSION	-	1.00	1.30	1.30	model version selector	
Basic device parameters						
TYPE	-	-1	1	1	1 for nMOS and -1 for pMOS	
TFOX	m	0 <	3.5 n		front oxide thickness	Fig. 2
TBOX	m	0 <	110 n		buried oxide thickness	Fig. 2
TSOI	m	0 <	50 n		silicon film thickness	Fig. 2
XJ	m	0 <	50 n		impurity doping depth	(255)
XLD	m	0 ≤	0	≤ 50 n	gate overlap length	(3)
XWD	m	-10 n ≤	0	≤ 100 n	gate overlap width	(5)
XLDC	m	-10 n ≤	0	≤ 100 n	XLD for capacitance	(4)
XWDC	m	-10 n ≤	0	≤ 100 n	XWD for capacitance	(6)
TPOLY	m	0 ≤	0		height of poly-Si gate for fringing cap	
NSUBS	cm ⁻³	1 · 10 ¹⁴ ≤	3 · 10 ¹⁷	≤ 1 · 10 ¹⁹	SOI layer impurity concentration	Fig. 2
NSUBB	cm ⁻³	0 <	4 · 10 ¹⁴		substrate impurity concentration	Fig. 2
NSUBP	cm ⁻³	1 · 10 ¹⁶ ≤	1 · 10 ¹⁷	≤ 1 · 10 ¹⁹	max pocket concentration	(100)
VFBC	V	-1.2 ≤	-1	≤ -0.8	flat-band voltage	(10)
VBI	V	1 ≤	1.1	≤ 1.2	built-in potential	(41)
smoothing coefficient between linear and saturation region						
DDLTMAX	-	0 ≤	10	≤ 20	smoothing coefficient for V_{ds}	(25)
DDLTSLP	μm^{-1}	0 ≤	10	≤ 20	L_{gate} -dependence of smoothing	(26)
DDLTICT	-	-3 ≤	0	≤ 20	coefficient	
SUBDLT	-		2 · 10 ⁻³		smoothing parameter (FB only)	
Mobility and velocity						
MUECB0	cm ² V ⁻¹ s ⁻¹	100 ≤	300	≤ 100 k	Coulomb scattering	(77)
MUECB1	cm ² V ⁻¹ s ⁻¹	15 ≤	30	≤ 10 k	Coulomb scattering	(77)
MUEPH0	-	0.25 ≤	0.3	≤ 0.3	phonon scattering	(78)
MUEPH1	cm ² V ⁻¹ s ⁻¹	2 k ≤	25 k	≤ 30 k	phonon scattering	(87)
	(V cm ⁻¹) ^{MUEPH0}					
MUETMP	-	0.5 ≤	1.5	≤ 2	temperature-dependence of phonon scattering	(142)
MUEPHL*	-		0		L-dependence of phonon mobility	
MUEPLP*	-		1		reduction	(87)
MUESR0	-	1.8 ≤	2	≤ 2.2	surface-roughness scattering	(88)
MUESR1	cm ² V ⁻¹ s ⁻¹	1 · 10 ¹⁴ ≤	2 · 10 ¹⁵	≤ 1 · 10 ¹⁶	surface-roughness scattering	(79)
	(V cm ⁻¹) ^{MUESR0}					
MUESRL*	-		0		L-dependence of surface roughness	(88)
MUESLP*	-		1		on mobility	
NDEP	-	0 ≤	1	≤ 1	depletion charge contribution to E_{eff}	(85)
NINV	-	0 ≤	0.5	≤ 1	inversion charge contribution to E_{eff}	(86)
NINVD*	V ⁻¹	0 ≤	0.0		inversion charge parameter	(80)
BB	-		2 or 1		high-field-mobility degradation	(89)
VMAX	cm·s ⁻¹	1 · 10 ⁶ ≤	7 · 10 ⁶	≤ 2 · 10 ⁷	saturation velocity	(91)
VOVER	-	0 ≤	0.01	≤ 1	velocity overshoot effect parameter	(91)
VOVERP	-	0 ≤	0.1	≤ 2	L_{eff} -dependence of velocity overshoot	(91)
VTMP*	cm·s ⁻¹	-2 ≤	0	≤ 1	temperature-dependence of saturation velocity	(143)

Parameter	Unit	Min	Default	Max	Description	See
Short-channel effect						
PARL1	m	$0 \leq$	10 n	≤ 50 n	SOI SCE parameter	(42)
PARL2	m	$0 \leq$	10 n	≤ 50 n	depletion width of channel/contact junction	(41)
SC1	-	$0 \leq$	0	≤ 20	magnitude of short-channel effect	(41)
SC2	V^{-1}	$0 \leq$	0	≤ 2	V_{ds} -dependence of short-channel effect	(41)
SC3*	$m \cdot V^{-1}$	$0 \leq$	0	≤ 100 n	V'_{bs} dependence of short-channel effect	(41)
SCR1	-	$0 \leq$	0	≤ 5	parameter for SCE via BOX	(42)
SCR2	-	$0 \leq$	0	≤ 5	parameter for SCE via BOX	(42)
SCR3	-	$0 \leq$	0.23	≤ 1	parameter for SCE via BOX	(42)
SCP1	-	$0 \leq$	0	≤ 20	magnitude of short-channel effect due to pocket	(47)
SCP2	V^{-1}	$0 \leq$	0	≤ 2	V_{ds} -dependence of SCE due to pocket	(47)
SCP3*	$m \cdot V^{-1}$	$0 \leq$	0	≤ 100 n	V_{ds} -dependence of SCE due to pocket	(47)
LP	m	$0 \leq$	0	≤ 300 n	pocket penetration length	(47)
Poly-silicon gate depletion effect						
PGD1	V	$0 \leq$	0	≤ 50 m	strength of poly-depletion effect	(73)
PGD2	V	$0 \leq$	1	≤ 1.5	threshold voltage of poly-depletion effect	(73)
PGD4*	-	$0 \leq$	0	≤ 3	L_{gate} -dependence of poly-depletion effect	(73)
Quantum mechanical effect						
QME1	mV	$0 \leq$	0	≤ 300 n	V_{gs} -dependence of quantum mechanical effect	(75)
QME2	V	$0 \leq$	0	≤ 3		
QME3	m	$0 \leq$	0	≤ 800 p	minimum T_{FOX} modification	(75)
Channel-length modulation						
CLM1	-	$0.5 \leq$	0.7	≤ 1	hardness coefficient of channel/contact junction	(92)
CLM2	-	$2 \leq$	Eq. (95)		coefficient for Q_B contribution	(94)
CLM3	-	$1 \leq$	1	≤ 5	coefficient for Q_I contribution	(94)
CLM5	-	$0 \leq$	1	≤ 5	CLM parameter	(96)
CLM6	-	$0 \leq$	0	≤ 5	CLM parameter	(96)
Punchthrough						
PTL	V^{1-PTP}	$0 \leq$	0		punchthrough parameter	(60)
PTLP	-		1		punchthrough parameter	(60)
PTP	-	$3 \leq$	3.5	≤ 4	punchthrough parameter	(60)
PT2	V^{-1}	$0 \leq$	0		punchthrough parameter	(60)
PT4	-	$0 \leq$	0		punchthrough parameter	(60)
PT4P	-	$0 \leq$	1		punchthrough parameter	(60)
NJUNC	$1/cm^3$	$0 <$	10^{20}		deep punchthrough parameter	(70)
XJPT	m	$0 <$	TSOI	$1.2 \times \mathbf{TSOI}$	deep punchthrough parameter	(66)
MUPT	$m^2/V/s$	$0 <$	0		deep punchthrough parameter	(62)
VFBPT	V		0		flatband offset for deep punchthrough effects	(71)
PSLIMPT	V		0		limiter for deep punchthrough effects	p.17
*GDL	-	0	0	220m	strength of high-field effect	(72)
*GDLP	-		0		modification of channel conductance	(72)
*GDLD	m		0		modification of channel conductance	(72)

Parameter	Unit	Min	Default	Max	Description	See
Narrow channel effects						
WFC	$\text{F}\cdot\text{cm}^{-2}\cdot\text{m}^{-1}$	$-5\cdot 10^{-15} \leq$	0	$\leq 1\cdot 10^{-6}$	Threshold voltage change due to cap change	(98)
WVTH0*	$\text{V}\cdot\mu\text{m}$		0		threshold voltage shift	(97)
NSUBCW**	-		0		substrate concentration modifier	(101)
NSUBCWP**	-		1		substrate concentration modifier	(101)
NSUBCMAX**	cm^{-3}		$5\cdot 10^{18}$		upper limit of substrate concentration	(101)
NSUBP0*	-		0		modification of pocket concentration	(100)
NSUBWP*	-		1		for narrow W	
NSUBCL**	-		0		substrate concentration modifier	(101)
NSUBCLP**	-		1		substrate concentration modifier	(101)
MUEPHW*	-		0		phonon-related mobility reduction	(102)
MUEPWP*	-		1		reduction	
MUESRW*	-		0		surface roughness-related mobility change	(103)
MUESWP*	-		1		surface roughness-related mobility change	(103)
WL2	-		0		threshold voltage shift due to	(115)
WL2P	-		1		small-size effect	
MUEPHS*	-		0		mobility change due to small size	(117)
MUEPSP*	-		1		mobility change due to small size	(117)
VOVERS*	-		0		modification of max velocity due to small size	(118)
VOVERS*	-		1		modification of max velocity due to small size	(118)
STI effects						
VTHSTI*	V		0		threshold voltage shift due to STI	(107)
VDSTI*	-		0		V_{ds} dependence of STI subthreshold	(107)
SCSTI1*	-		0		the same effect as SC1 but at STI edge	(110)
SCSTI2*	V^{-1}		0		the same effect as SC2 but at STI edge	(110)
NSTI	cm^{-3}	$1\cdot 10^{16} \leq$	$5\cdot 10^{17}$	$\leq 1\cdot 10^{19}$	substrate impurity concentration at STI edge	(105)
WSTI	m		0		width of high-field region at STI edge	(113)
WSTIL*	-		0		channel-length dependence of WSTI	(114)
WSTILP*	-		1.0		channel-length dependence of WSTI	(114)
WSTIW*	-		0		channel-width dependence of WSTI	(114)
WSTIWP*	-		1.0		channel-width dependence of WSTI	(114)
WL1	m		0		small-size effect parameter for STI leakage	(111)
WL1P	-		1.0		small-size effect parameter for STI leakage	(111)
NSUBCSTI1	m		0		channel concentration modifier	(121)
NSUBCSTI2	-		0		channel concentration modifier	(120)
NSUBCSTI3	-		1.0		channel concentration modifier	(121)
NSUBPSTI1	m		0		pocket concentration modifier	(127)
NSUBPSTI2	-		0		pocket concentration modifier	(125)
NSUBPSTI3	-		1		pocket concentration modifier	(127)
MUESTI1	m		0		mobility change due to diffusion length	(131)
MUESTI2	-		0		mobility change due to diffusion length	(130)
MUESTI3	-		1		mobility change due to diffusion length	(131)
SAREF	m		$1\cdot 10^{-6}$			p. 26
SBREF	m		$1\cdot 10^{-6}$			p. 26
Temperature dependence						
EG0	eV	$1 \leq$	1.1785	≤ 1.3	bandgap	(140)
BGTMP1	$\text{eV}\cdot\text{K}^{-1}$	$50\mu \leq$	90.25μ	$\leq 100\mu$	temperature dependence of bandgap	(139)
BGTMP2	$\text{eV}\cdot\text{K}^{-2}$	$-1\mu \leq$	0.1μ	$\leq 1\mu$	temperature dependence of bandgap	(139)
TNOM	$^{\circ}\text{C}$	$22 \leq$	27	≤ 32	temperature selected as a nominal value	(139)

Parameter	Unit	Min	Default	Max	Description	See
Parasitic resistances						
RSH	$\Omega/\square$	$0 \leq$	80μ	$\leq 10 \text{ m}$	source/drain sheet resistance	(157)
RSHG	$\text{V A}^{-1}\text{square}$		0		gate sheet resistance	(162)
RDRDJUNC	[m]		1.0×10^{-6}		junction depth at channel/drift region	(156)
RDRBB	[—]		1.0		high field mobility in drift region (drain)	(149)
RDRBBS	[—]		1.0		high field mobility in drift region (source)	(150)
RDRMUE	$[\text{cm}^2(\text{V} \cdot \text{s})^{-1}]$	100	1k	3k	mobility in drift region (drain)	(144)
RDRMUES	$[\text{cm}^2(\text{V} \cdot \text{s})^{-1}]$	100	1k	3k	mobility in drift region (source)	(145)
RDRMUEL	[—]		0		mobility in drift region	(154)
RDRMUELP	[—]		1		L_{gate} dependence	(154)
RDRBBTMP	[1/K]		0		temp. dependence of RDRBB , S	(149)(150)
RDRMUETMP	[—]	0	0	2.0	temp. dependence of RDRMUE , S	(144)(145)
RDRVMAX	$[\text{cm s}^{-1}]$	1MEG	30MEG	100MEG	saturation velocity in drift region	(147)
RDRVMAXL	[—]		0		saturation velocity in drift region	(155)
RDRVMAXLP	[—]		1		L_{gate} dependence	(155)
RDRVMAXW	[—]		0		saturation velocity in drift region	(155)
RDRVMAXWP	[—]		1		W_{gate} dependence	(155)
RDRVMAXS	$[\text{cm s}^{-1}]$	1MEG	30MEG	100MEG	saturation velocity in source region	(148)
RDRVTMP	[—]	-2.0	0	1.0	temp. dependence of RDRVMAX	(147)(148)
LDRIFT	[m]	0	1.0×10^{-6}		length of drift region (drain)	Fig.3
LDRIFTS	[m]	0	1.0×10^{-6}		length of drift region (source)	Fig.3
RBULK0	[Ohm]		0		body resistance (COBCNODE =1)	(164) Fig.13.4
RBULKW	[Ohm/m]		0		body resistance coefficient (COBCNODE =1)	(164) Fig.13.4
RBSB	[Ohm]		50		body resistance (source side) (CORBNET =1)	Fig.9
RBDB	[Ohm]		50		body resistance (drain side) (CORBNET =1)	Fig.9
Capacitance						
XQY	m	$0 \leq$	0	$\leq 50 \text{ n}$	distance from junction to max field point	(166)
XQY1	$\text{F} \cdot \mu\text{m}^{\text{XQY2}-1}$	$0 \leq$	0		V'_{bs} -dependence of Q_y	(166)
XQY2	-	$0 \leq$	2		L_{gate} -dependence of Q_y	(166)
LOVER	m	$0 <$	30 n		drain overlap length	(169)
NOVER	cm^{-3}		$1 \cdot 10^{19}$		impurity concentration in overlap region	(169)
VFBOVER	V		0		flat-band voltage in overlap region	(170)
OVSLP	m V^{-1}		$2.1 \cdot 10^{-7}$		coefficient for overlap capacitance	(171)
OVMAg	V		0.6		coefficient for overlap capacitance	(171)
CGDO	$\text{F} \cdot \text{m}^{-1}$	$0 \leq$	0	$\leq 10^2 \text{ nm} \cdot C_{\text{FOX}}$	gate-to-drain overlap cap	Section 14.2
CGSO	$\text{F} \cdot \text{m}^{-1}$	$0 \leq$	0	$\leq 10^2 \text{ nm} \cdot C_{\text{FOX}}$	gate-to-source overlap cap	Section 14.2
CGBO	$\text{F} \cdot \text{m}^{-1}$	$0 \leq$	0		gate-to-bulk overlap cap	(177)

Parameter	Unit	Min	Default	Max	Description	See
Substrate current						
VFBSUB	V		-1		flatband voltage for I_{sub} calculation (184)(185)(190)(191)	
VFBSUBL	-		0		L_{gate} -dependence of VFBSUB (184)(185)(190)(191)	
VFBSUBLP	-		1		L_{gate} -dependence of VFBSUB (184)(185)(190)(191)	
SUB1	V^{-1}		0.01		substrate current coefficient of magnitude	(181)
SUB1L	m		$2.5 \cdot 10^{-3}$		L_{gate} -dependence of SUB1	(181)
SUB1LP	-		1		L_{gate} -dependence SUB1	(181)
SUB2	V		20		substrate current coefficient of exponential term	(181)
SUB2L	m	$0 \leq$	$2 \cdot 10^{-6}$	≤ 1	L_{gate} -dependence of SUB2	(181)
SVDS	-		3		Substrate current dependence on V_{ds} (182)(183)(187)(188)	
SLG	-		$3 \cdot 10^{-8}$		substrate current dependence on L_{gate}	(187)(188)
SLGL	-		0		substrate current dependence on L_{gate}	(187)(188)
SLGLP	-		1		substrate current dependence on L_{gate}	(187)(188)
SVGS	-		0.8		substrate current dependence on V_{gs}	(189)
SVGSL	-		0		L_{gate} -dependence of SVGS	(189)
SVGSLP	-		1		L_{gate} -dependence of SVGS	(189)
SVGSW	-		0		W_{gate} -dependence of SVGS	(189)
SVGSWP	-		1		W_{gate} -dependence of SVGS	(189)
SVBS	-		0.5		substrate current dependence on V'_{bs}	(186)(192)
SVBSL	-		0		L_{gate} -dependence of SVBS	(186)(192)
SVBSLP	-		1		L_{gate} -dependence of SVBS	(186)(192)
IBPC1	$\text{V} \cdot \text{A}^{-1}$	$0 \leq$	0	$\leq 1 \cdot 10^{12}$	impact-ionization-induced bulk - potential change	(194)
IBPC2	V^{-1}	$0 \leq$	0	$\leq 1 \cdot 10^{12}$		
Gate leakage current						
GLEAK1	$\text{V}^{-3/2} \cdot \text{s}^{-1}$		$1 \cdot 10^4$		gate-to-channel current coefficient	(15.2)
GLEAK2	$\text{V}^{-1/2} \cdot \text{m}^{-1}$		20MEG		gate-to-channel current coefficient	(15.2)
GLEAK3	-		$3 \cdot 10^{-1}$		gate-to-channel current coefficient	(196)
GLEAK4	m^{-1}		4		gate-to-channel current coefficient	(196)
GLEAK5	$\text{V} \cdot \text{m}^{-1}$		$7.5 \cdot 10^3$		gate-to-channel short channel correction	(196)
GLEAK6	V		$2.5 \cdot 10^{-1}$		gate-to-channel V_{ds} -dependence correction	(15.2)
GLEAK7	m^2		$1 \cdot 10^{-6}$		gate-to-channel L and W dependence correction	(15.2)
GLKSD1	$\text{A} \cdot \text{m} \cdot \text{V}^{-2}$		1 f		gate-to-S/D current coefficient	(204)
GLKSD2	$\text{V}^{-1} \cdot \text{m}^{-1}$		5 MEG		gate-to-S/D current coefficient	(204)
GLKSD3	m^{-1}		-5 MEG		gate-to-S/D current coefficient	(204)
GLKB1	$\text{A} \cdot \text{V}^{-2}$		$5 \cdot 10^{-16}$		gate-to-S/D current coefficient	(202)
GLKB2	$\text{m} \cdot \text{V}^{-1}$		1		gate-to-S/D current coefficient	(202)
GLKB3	V		0		gate-to-S/D current coefficient	(203)
FVBS	-	$0 \leq$	0		V'_{bs} dependence of Fowler-Nordheim current	(214)
EVB1	$\text{V}^{-2} \text{s}^{-1}$	$0 \leq$	0		electron tunneling from valence band	(213)
EVB2	$\text{V} \cdot \text{m}^{-1}$	$0 \leq$	0		electron tunneling from valence band	(213)
EVB3	-		0		electron tunneling from valence band	(214)
Gate-induced drain/source leakage						
GIDL1	$\text{A} \cdot \text{V}^{-3/2} \text{C}^{-1} \text{m}$		$5 \cdot 10^{-6}$		magnitude of GIDL	(208)
GIDL2	$\text{V}^{-2} \text{m}^{-1} \text{F}^{-3/2}$		$1 \cdot 10^6$		field-dependence of GIDL	(208)
GIDL3	-		0.3		V_{ds} -dependence of GIDL	(209)
GIDL4*	V		0		threshold for V_{ds} dependence	(209)
GIDL5*	-		0.2		high-field correction	(210)
GIDLVB	V^3		0.5		body bias dependence	(208)
Floating-body effect						
QHE1	-	$0 <$	1.5		FBE parameter	(217)
QHE2	V	$0 <$	0.35		FBE parameter	(217)
QHSMAX	Coulomb/ m^2		10^{-3}		(for body-tie devices only (COBCNODE=1))	p.45
HIST1	V	$0 <$	$1 \cdot 10^{-8}$		history-effect parameter	(224)
HIST2	A	$0 <$	$1 \cdot 10^{-20}$		history-effect parameter	(224)

Parameter	Unit	Min	Default	Max	Description	See
Source/bulk and drain/bulk diodes						
JS0	$\text{A}\cdot\text{m}^{-2}$	$0 <$	$1 \cdot 10^{-4}$		saturation current density	(227)
NJ	-		1		emission coefficient	(227)
XTI	-		2		temp coeff. for forward current densities	(227)
XTI2	-		0		temp coeff. for reverse current densities	(228)
DIVX	V^{-1}		0		reverse current coefficient	(238)
CJ	$\text{F}\cdot\text{m}^{-2}$		$5 \cdot 10^{-4}$		bottom junc cap/unit area at zero bias	(245)
CJSW	$\text{F}\cdot\text{m}^{-1}$		$5 \cdot 10^{-10}$		sidewall junc cap grading coefficient	(246)
CJSWG	$\text{F}\cdot\text{m}^{-1}$		$5 \cdot 10^{-10}$		sidewall junc cap per unit length at zero bias	(247)
MJ	-		0.33		bottom junc cap grading coefficient	p. 49
MJSW	-		0.33		sidewall junc cap grading coefficient	(260)
MJSWG	-		0.33		gate sidewall junc cap grading coefficient	(265)
PB	V		1		bottom junc built-in potential	(250)
PBSW	V		1		sidewall junc built-in potential	(258)
PBSWG	V		1		gate sidewall junc built-in potential	(263)
VDIFFJ	V	$0 <$	$1.6 \cdot 10^{-3}$		diode threshold voltage at junction	(235)
Body-tie contact						
XWDBT	m				BT overlap width	(227)
CBTBN	F/m^2				N^+ poly capacitance	(227)
CBTBP	F/m^2				P^+ poly capacitance	(227)
VFBTBP	V		0		flatband voltage for overlapped MOS-FET part	
1/f noise						
NFALP	$\text{cm}\cdot\text{s}$		$1 \cdot 10^{-19}$		contribution of mobility fluctuation	(304)
NFTRP	V^{-1}		$1 \cdot 10^{10}$		ratio of trap density to attenuation coeff.	(304)
CIT*	$\text{F}\cdot\text{cm}^{-2}$		0		cap caused by the interface trapped carriers	(305)
FALPH	-		1.0		exponent of frequency f as in $1/f^{\text{FALPH}}$	(304)
Non-quasi-static model						
DLY1	s		$1 \cdot 10^{-10}$		coeff. for delay due to diffusion of carriers	(316)
DLY2	-		0.7		coeff. for delay due to conduction of carriers	(317)
DLY3	Ω		$8 \cdot 10^{-7}$		coefficient for RC delay of bulk carriers	(319)
Self-heating						
RTH0	Kcm/W		0.1		Thermal resistance	p. 47
CTH0	Ws/(Kcm)		$1 \cdot 10^{-7}$		thermal capacitance	p. 47
Conservation of symmetry at $V_{\text{ds}} = 0$ for short-channel MOSFETs						
VZADD0	V		10m		symmetry conservation coefficient	
PZADD0	V		5m		symmetry conservation coefficient	
internal limiter						
VGSMIN	V		$-5 \cdot \text{TYPE}$		surface potential limiter (BT only)	

27 List of instance parameters, default values, and ranges

The maximum and minimum limits of the instance parameters are recommended values.

Parameter	Unit	Min	Default	Max	Description	See
L	m	$0 <$	$5 \cdot 10^{-6}$		Gate length (L_{gate})	(1)
W	m	$0 <$	$5 \cdot 10^{-6}$		Gate width (W_{gate})	(2)
AD	m^2	$0 \leq$	0		Area of drain junction	(232)
AS	m^2	$0 \leq$	0		Area of source junction	(239)
PD	m	$0 \leq$	0		Perimeter of drain junction	(232)
PS	m	$0 \leq$	0		Perimeter of source junction	(239)
NRD	m	$0 \leq$	0		Number of source squares	(151)
NRS	m	$0 \leq$	0		Number of drain squares	(157)
LDRIFT	m	$0 \leq$	1.0×10^{-6}		length of drift region (drain side)	Fig.3
LDRIPTS	m	$0 \leq$	1.0×10^{-6}		length of drift region (source side)	Fig.3
XGL	m	$0 \leq$	0		Offset of gate length	(162)
XGW	m	$0 \leq$	0		Distance from gate contact to channel edge	(162)
M	-	$1 \leq$	1		Multiplication factor	p. 61
NF	-	$1 \leq$	1		Number of gate fingers	(5)
NGCON	-	$1 \leq$	1		Number of gate contacts	(162)
SA	m	$0 \leq$	0		Length of diffusion between gate and STI	p. 26
SB	m	$0 \leq$	0		Length of diffusion between gate and STI	p. 26
SD	m	$0 \leq$	0		Length of diffusion between gate and gate	p. 26
RBDB	Ohm	$0 \leq$	50		Body resistance (drain side)	p. 33
RBSB	Ohm	$0 \leq$	50		Body resistance (source side)	p. 33
PDBCP	m	$0 \leq$	0		Parasitic perimeter length for body contact at drain	Sec.20.1.2
PSBCP	m	$0 \leq$	0		Parasitic perimeter length for body contact at source	Sec.20.1.2
NBT			1		Number of body contact	p. 53
LBT	m	$0 \leq$	0		Length of gate over body-contact well	p. 53
WBTN	m	$0 \leq$	0		Distance of gate protrusion over body-contact well	p. 53
WBTP	m	$0 \leq$	0		Distance of gate protrusion over body-contact well	p. 53
ABTN	m^2		0		Area of N^+ poly area of body contact	p. 53
ABTP	m^2		0		Area of P^+ poly area of body contact	p. 53
TEMP	$^{\circ}\text{C}$		27		Device temperature	(134)
DTEMP	$^{\circ}\text{C}$		0		Device temperature change	(134)

28 Overview of the Parameter-Extraction Procedure

In HiSIM, device characteristics are strongly dependent on basic device parameter values, such as the impurity concentration and the oxide thickness. Therefore, the parameter-value extraction has to be repeated with measured characteristics of different devices in a specific sequence until extracted parameter values reproduce all device characteristics consistently and reliably. To achieve reliable results, it is recommended to start with initial parameter values according to the recommendations listed in the table below. Since some of the model parameters such as T_{FOX} are difficult to extract, they are expected to be determined directly by dedicated measurements. Threshold voltage measurements allow to derive a rough extraction for the model parameters referred to as “basic device parameters”. The parameters identified with the symbol “*” in the Model Parameter Table are initially fixed to zero.

Determined by dedicated measurements (not changed during extraction procedure)	Default values listed in the section 26.1 are used initially for the groups of parameters listed below
TFOX TBOX TSOI	basic device parameters (not listed on left side) velocity quantum effect substrate current gate leakage GIDL mobility source/bulk and drain/bulk diodes noise subthreshold swing non-quasi-static model overlap capacitances

The sequence of device selection for the parameter extraction is recommended in 4 steps

1. Long-channel devices
2. Short-channel devices
3. Long-narrow devices
4. Short-narrow devices

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